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Harnessing Artificial Intelligence's Potential in the Middle East and Central Asia

Prepared by Abdullah AlHassan, Serpil Bouza, Vrinda Handa, Giovanni Melina, Nareg Mesrobian, Karmen Naidoo, John Ralyea, Subi Velkumar, and Philippe Wingender

With contributions from Leen Aghabi, Sana Almunizel, Gregorio Impavido, and Yuan Gao Rollinson

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Executive Summary

Artificial intelligence (AI) represents a meaningful opportunity to raise productivity, strengthen medium-term growth, and support economic diversification across the Middle East, North Africa, Afghanistan, and Pakistan (MENAP) and the Caucasus and Central Asia (CCA). While the war in the Middle East that began in February 2026 has elevated geopolitical uncertainty and poses new risks to the region's economic trajectory, it also reinforces the strategic importance of AI-driven diversification, particularly for hydrocarbon-dependent economies seeking to reduce reliance on oil and gas revenues, enhance resilience, and mitigate potential long-term economic scarring.

AI adoption potential differs markedly across countries. This paper assesses three crucial dimensions for adoption: AI exposure (the extent to which occupations and industries are likely to be affected by AI); AI preparedness (the extent to which countries have the infrastructure, skills, and institutions needed to integrate AI); and AI access (the availability of AI-specific technologies).

Differences across these dimensions imply heterogeneous growth outcomes. Model-based simulations suggest that AI adoption could raise real GDP growth by 0.1–0.6 percentage points per year, depending on the country and scenario. Growth gains in the Gulf Cooperation Council (GCC) economies could be about 1.5 times larger than in CCA economies and roughly double those in MENAP low-income countries, reflecting higher AI exposure of employment and sectors, stronger preparedness, and better access to advanced technologies.

Several GCC economies have been pursuing large-scale investments aimed at positioning themselves as AI service hubs, building on structural advantages, including abundant low-cost energy. While the war in the Middle East may dent investor confidence and the area's safe-haven appeal, with potential implications for economic agendas going forward, it has also reinforced the strategic case for AI investments. Accelerating diversification away from hydrocarbons has become more urgent, and AI-led transformation offers a path to new sources of growth. Provided AI infrastructure investment continues, its payoff increases if AI services exports expand. Under stronger export growth, gains from AI-related investment could roughly double relative to a conservative export baseline. Realizing this upside, however, hinges on sustained peace and stability, improvements in AI preparedness and adoption capacity in importing countries, as well as regulatory arrangements with advanced economies.

Countries without ambitions to become global AI hubs can still benefit from AI adoption. Reforms that raise AI preparedness could add up to 0.1 percentage point to annual growth over a decade. Although structural differences in AI adoption are likely to persist in the medium term, improving readiness can help reduce potential growth differentials. At the same time, AI readiness upgrades are not a substitute for broader structural reforms. Governance improvements, trade openness, and private sector development remain essential.

AI adoption also introduces risks that require proactive mitigation. In the near term, highly leveraged AI-related investment in emerging hubs, especially the GCC, creates financial stability risks. If productivity gains from AI are less than anticipated, large-scale infrastructure could become underutilized, leading to stranded digital assets and weaker returns. These risks may be amplified by war-related uncertainty and shifts in risk appetite, and they underscore the importance of integrating AI-specific scenarios in macro-financial policy frameworks.

Over the medium term, AI-driven automation and skill-biased technological change could intensify labor market pressures. With 30–50 percent of employment exposed to AI across MENAP and CCA economies, job reallocation, skill mismatches, and inequality risks may rise. Women, who tend to work in occupations with higher AI exposure but also stronger complementarity potential, face both risks and opportunities. Younger workers, often concentrated in entry-level roles with lower complementarity, may be particularly vulnerable in economies with already high youth unemployment. These risks strengthen the case for well-targeted social protection systems and active labor market policies to support workers through the transition and help them adapt to shifting skill requirements.

Acronyms and Abbreviations

AE	advanced economy
AI	artificial intelligence
AIPI	Artificial Intelligence Preparedness Index
CCA	Caucasus and Central Asia
EM	emerging market
EMMI	emerging market and middle-income economy
GCC	Gulf Cooperation Council
ICT	information and communication technology
IT	information technology
LIC	low-income country
MENAP	Middle East, North Africa, Afghanistan, and Pakistan
TFP	total factor productivity

1. Introduction

Artificial intelligence (AI) offers the Middle East, North Africa, Afghanistan, and Pakistan (MENAP) and the Caucasus and Central Asia (CCA) regions a unique opportunity to accelerate productivity, strengthen medium-term growth prospects, and support economic diversification. This opportunity is especially salient in a region where productivity growth lagged global peers for decades, nonresource sectors remain underdeveloped, and many economies face rising pressure to improve competitiveness in a more fragmented and risk-prone global landscape. Some countries in the region retain structural advantages for AI deployment—abundant low-cost energy for data centers (including renewable capacity), strategic geographic location, relationships with leading global technology firms, and national strategies that facilitate top-down implementation of AI-friendly regulation and infrastructure (Albous, Al-Jayyousi, and Stephens 2025). These factors could support the emergence of regional hubs for data-intensive industries, cloud services, and AI-enabled logistics. In particular, some Gulf Cooperation Council (GCC) countries—especially the United Arab Emirates and Saudi Arabia—have been rapidly scaling up investment in data centers and cloud infrastructure, with ambitions to become AI service hubs as part of broader diversification toward higher value-added sectors. While the war in the Middle East that began in February 2026 has heightened uncertainty, disrupted energy and trade routes, and tested the region's safe-haven appeal, it has also reinforced the rationale for GCC's large-scale AI investments. The need to reduce reliance on hydrocarbon exports and contain the potential longer-term economic scarring from the war has intensified, raising the expected payoff of AI-led diversification.

Realizing AI's potential is not automatic. The IMF's Artificial Intelligence Preparedness Index (AIPI)[1] highlights pronounced disparities across MENAP and CCA economies in digital infrastructure, human capital, governance, and innovation ecosystems. Although some economies have progressed in integrating AI into production and public services, many remain at early stages of adoption, producing a highly uneven regional landscape. These gaps risk turning AI into a new fault line, amplifying existing productivity differentials and widening income disparities across regions. Closing the AI divide is therefore essential for raising growth and supporting competitiveness.

The challenges are macro critical not only for economies lagging in adoption but also for those aspiring to become AI hubs. Returns on large-scale AI-specific infrastructure investments depend on sufficiently broad adoption at the domestic, regional, and global levels. Without sufficient demand, even world-class AI hubs may face underutilized capacity and the risk of stranded digital assets. Moreover, significant investment in AI, which is often cofinanced by sovereign states and major multinational technology firms, could expose regional economies to boom-bust dynamics if global valuations of AI assets correct sharply (IMF 2026).

Beyond near-term financial stability considerations, medium-term risks depend on the pace and breadth of AI diffusion within countries. A key concern is labor market adjustment: job reallocation, shifting skill requirements, and potential displacement in some occupational groups. This raises the risk of higher inequality and underscores the need for inclusive, well-managed transitions, particularly in the MENAP region, where youth unemployment is already high.

[1] The AIPI measures the level of AI readiness across 174 countries using a comprehensive set of macrostructural indicators covering four dimensions: digital infrastructure, human capital and labor market policies, innovation and economic integration, and regulation and ethics. This index draws on official statistics, surveys of hard data, and perception-based surveys compiled by eight institutions: the Fraser Institute, International Labour Organisation (ILO), International Telecommunications Union (ITU), United Nations, United Nations Conference on Trade and Development (UNCTAD), the Universal Postal Union (UPU), World Bank, and World Economic Forum (WEF).

This paper analyzes the macroeconomic impact of AI in the MENAP and CCA regions along four dimensions:

- First, it adapts the global general equilibrium model analysis of Cerutti and others (2025) to the regions of interest to examine the uneven economic effects of AI adoption. The analysis emphasizes that AI-driven gains are expected to be heterogeneous across countries and subregions, reflecting differences in sectoral exposure to AI, preparedness to integrate these technologies, and access to essential inputs such as data, cloud services, and frontier technologies.
- Second, it assesses the emergence of AI hubs in the GCC and the macroeconomic implications of large AI-related infrastructure investment, with attention to expected domestic and foreign demand for AI services.
- Third, it examines alternative macroeconomic scenarios reflecting uncertainty around AI productivity and diffusion, including downside scenarios with restricted access to AI-specific infrastructure and upside scenarios where policy reforms raise AI preparedness.
- Fourth, it discusses the challenges AI poses to MENAP and CCA economies, including short-term risks stemming from a potential correction in market valuations of AI prospects and medium-term risks that may emerge as adoption progresses, particularly in labor markets.

The paper contributes to a rapidly growing literature on the macroeconomic implications of AI. Much of the existing research, including several IMF contributions, focuses on labor market effects. A large body of work adopts task-based approaches to measure occupational exposure to AI and assess how AI-driven productivity gains may affect employment and wages (see, among others, Webb 2020; Felten, Raj, and Seamans 2021, 2023; Pizzinelli and others 2023; Cazzaniga and others 2024; Eloundou and others 2024; Korinek 2024; Korinek and Stiglitz 2025; Rockall, Tavares, and Pizzinelli 2025; Handa and others 2025). Some studies provide estimates of AI's labor market effects in the United States (see, for example, Pizzinelli and others 2023; Huang 2024; Brynjolfsson, Chandar, and Chen 2025; Hosseini and Lichtinger 2025; Liu, Wang, and Yu 2025). Although most empirical evidence focuses on advanced economies (AEs), recent cross-country studies point to substantial heterogeneity in AI's labor market and macroeconomic effects across regions and demographic groups (see OECD 2023; Briggs and Kodnani 2023; Gmyrek, Berg, and Bescond 2023; Albanesi and others 2023; Dorville, Filippucci, and Marcolin forthcoming; Korinek and Juelfs 2023; Brollo and others 2024; Berg and others 2025; Cerutti and others 2025).

At the macroeconomic level, however, there is less consensus regarding the magnitude of potential productivity gains. Focusing on the United States, Acemoglu (2025) cautions that productivity gains may be modest once more complex, context-dependent tasks are considered and emphasizes possible inequality implications. By contrast, Aghion and Bunel (2024) identify channels through which AI could substantially boost productivity gains, both through task-level improvements and idea generation. Whereas these studies largely rely on single-country frameworks, Cerutti and others (2025) integrate earlier insights related to AI exposure, preparedness, and access within a multicountry, multisector model to offer scenarios on global impacts and regional differences for AEs, emerging markets (EMs), and low-income countries (LICs) related to AI adoption.

Despite this expanding literature, region-wide macroeconomic estimates for the MENAP and CCA regions remain limited. Existing studies are either global in scope or focused on selected countries. For example, PricewaterhouseCoopers (2018) estimates AI's potential contribution to GDP for a subset of countries but does not assess growth effects or subregional heterogeneity. A recent World Bank (2025) report focuses on digital and AI developments in the GCC but does not quantify their medium-term growth effects.

In this context, this paper advances the literature along two main dimensions. First, it provides the first systematic, scenario-based quantification of AI's macroeconomic effects at the subregional level across MENAP and CCA economies, explicitly capturing heterogeneity in exposure, preparedness, and access. By

adapting the multicountry general equilibrium framework of Cerutti and others (2025) to these regions, the paper estimates medium-term impacts on output, inflation, external balances, and exchange rates under alternative AI productivity scenarios.

Second, the paper introduces a novel focus on strategic complementarities between emerging AI hubs—primarily in the GCC—and potential AI adopters in the rest of MENAP. It analyzes how returns on large-scale AI infrastructure investment depend critically on broad and productive AI adoption both within MENAP (beyond the GCC) and globally. By conducting scenario analysis in a general equilibrium model, the study contributes to a still-limited macroeconomic literature examining AI's implications for external balances, exchange rates, and inflation and offers policy guidance to help maximize gains while mitigating risks.

The remainder of the paper is organized as follows. Section 2 documents heterogeneity in AI exposure, preparedness, and access—the main drivers of AI adoption—across the MENAP and CCA regions. Section 3 reviews the evolving AI investment landscape, particularly in the GCC, and discusses potential sources of demand for AI services and implications for diversification. Section 4 presents model-based estimates of the macroeconomic impact of AI productivity shocks under baseline and alternative scenarios, including enhanced preparedness and restricted access. Section 5 examines macrofinancial risks and outlines key policy challenges. Section 6 concludes with policy implications. The paper also includes three country boxes—Jordan, Kazakhstan, and Saudi Arabia (Boxes 3–5)—providing selected examples of national AI strategies.

2. The Region's AI Adoption Landscape

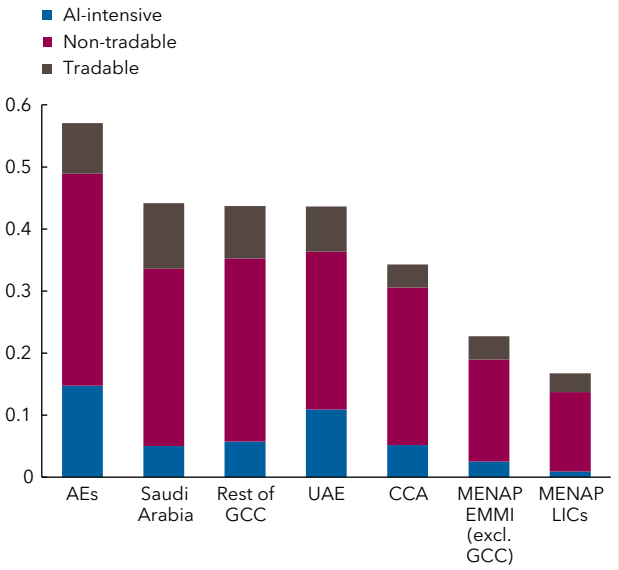
This section assesses the three dimensions of AI adoption: AI exposure (the extent to which occupations and industries are likely to be affected by AI); AI preparedness (the extent to which countries have appropriate regulatory frameworks, governance structures, institutions, digital infrastructure, human capital, and labor market policies); and AI access (the availability of AI-specific technologies). Reducing disparities across these dimensions in the MENAP and CCA regions would strengthen countries' ability to harness AI's benefits.

A. AI Exposure

An economy's AI exposure—the extent to which AI can augment or substitute for workers and production processes—reflects its occupational and skill distribution and its sectoral composition.[2] This paper applies established methodologies to map the overlap between AI capabilities and the abilities required in each occupation.[3] In general, occupations most exposed to AI require higher skills (for example, managers and professionals), where a large share of tasks may be complementary to AI adoption. Industries with a relatively high share of cognitive-intensive occupations—such as pharmaceuticals; computer, electronic, and optical equipment; air transport; telecoms; IT; finance; and professional and scientific activities—are therefore considered AI-intensive (Annex Table 1.3). Some associate professional and clerical occupations also overlap strongly with AI capabilities but may face higher displacement risk (for example, roles such as engineering technicians, film and video editors, and telemarketers, among others).

Employment in the MENAP and CCA regions is less exposed to AI than in AEs (Figure 1). Saudi Arabia, the United Arab Emirates, and other GCC economies have the highest shares of highly exposed jobs within MENAP and CCA economies, at almost 45 percent on average. Given the relatively small share of the AI-intensive sector in GCC economies, much of this exposure reflects occupations in non-tradable sectors (including many

Figure 1. Labor Market AI Exposure by Sector
(Share of employment)



Sources: Cazzaniga and others (2024); and IMF staff calculations.
Note: AEs = advanced economies; CCA = Caucasus and Central Asia; EMMIs = emerging market and middle-income economies (excluding GCC); GCC = Gulf Cooperation Council; LIC = low-income countries; MENAP = Middle East, North Africa, Afghanistan, and Pakistan; UAE = United Arab Emirates.

[2] Occupations serve as the primary channel through which AI impacts industries and economies. To study these effects, it is standard to conceptualize individual occupations as a bundle of tasks and to consider which tasks can be replaced or complemented by technology (Cazzaniga and others 2024; Cerutti and others 2025). We use detailed occupation- and industry-level data from the ILO to compute AI exposure (see Annex Table 1.1). Given the level of informality in MENAP and CCA, the same occupation may have a different task distribution relative to AEs; therefore, the calculated AI exposures presented here can be interpreted as an upper bound.

[3] The AI exposure measure follows Felten, Raj, and Seamans (2021, 2023); Pizzinelli and others (2023); and Cazzaniga and others (2024). AI exposure as constructed by Felten, Raj, and Seamans (2021, 2023) is based on 10 selected AI capabilities that capture the most likely and common uses of AI: abstract strategy games, real-time video games, image recognition, visual question answering, image generation, reading comprehension, language modeling, translation, speech recognition, and instrumental track recognition. Importantly, although newer applications (for example, agent-based workflows) are not explicitly captured, they largely build on the same underlying task content (language modeling, reading comprehension, and speech and image recognition).

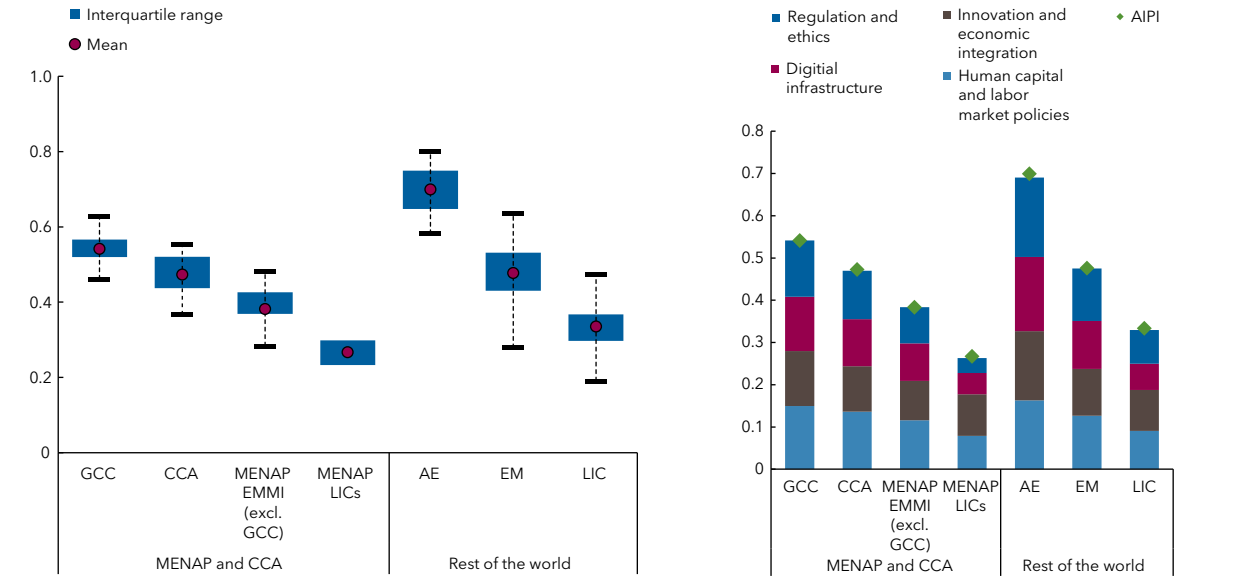
services), followed by the tradable sectors (including agriculture, mining, and manufacturing). The United Arab Emirates is an exception: About one-quarter of AI-exposed employment is in the AI-intensive sector, similar to the share found in AEs. MENAP EMMIs (excluding the GCC) and LICs have lower AI exposure, at about 23 percent and 17 percent of employment, respectively. In the CCA economies, overall exposure is close to 35 percent, with a notable portion concentrated in AI-intensive sectors.

B. AI Preparedness

AI preparedness varies widely across the MENAP and CCA regions (Figure 2, panel 1). The AIPI benchmarks 174 countries using four key dimensions: digital infrastructure, human capital, technological innovation, and legal frameworks. Within the MENAP and CCA regions, GCC countries—especially Saudi Arabia and the United Arab Emirates—score highest, exceeding the average for EMs, though still below advanced-economy levels. On average, the GCC's preparedness is about 1.5 times that of MENAP EMMIs (excluding the GCC) and about double that of MENAP LICs. The CCA region is broadly in line with the EM average, whereas MENAP EMMIs lag behind. MENAP LICs remain below the global LIC average, reflecting that many are fragile and conflict-affected states.

Disaggregated AIPI components help identify policy priorities to bridge these disparities (Figure 2, panel 2). GCC countries are closer to AEs in human capital and labor market policies than in digital infrastructure, innovation, and regulation. The CCA region broadly mirrors the EM average, whereas MENAP EMMIs (excluding the GCC) trail the EM average across all dimensions. MENAP LICs lag the global LIC average across nearly all dimensions. Across both regions, weaknesses in regulation and ethics—often rooted in broader governance gaps—are the principal shortfall.

Figure 2. AI Preparedness



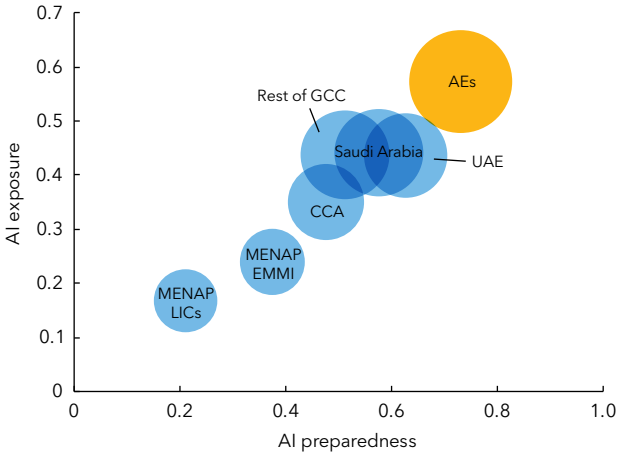
Sources: Cazzaniga and others (2024); and IMF staff calculations.

Note: AE = advanced economy; AI = Artificial Intelligence; AIPI = Artificial Intelligence Preparedness Index; CCA = Caucasus and Central Asia; EM = emerging market (includes both oil importers and oil exporters); EMMIs = emerging market and middle-income economies; GCC = Gulf Cooperation Council; LIC = low-income country; MENAP = Middle East, North Africa, Afghanistan, and Pakistan.

There is a positive correlation between the overall level of AI exposure and preparedness (Figure 3). For higher-income countries (particularly the GCC), higher AI exposure implies more immediate risks, including to

employment, but it also means that there are greater opportunities because employment and sectoral structures are more conducive to adoption, and preparedness is higher. In addition, the larger share of AI exposure by GDP in the GCC countries (around 35–40 percent of GDP) underscores the potential for significant productivity and growth gains. In contrast, MENAP LICs have lower exposure in terms of both employment and the share of the economy that could benefit from AI productivity gains, along with weaker readiness, limiting near-term gains.

Figure 3. AI Exposure versus AI Preparedness
(Share of employment, index 0–1)



Sources: Cazzaniga and others (2024); and IMF staff calculations.
Note: Bubble size represents the share of GDP that is highly exposed to AI. AE = advanced economy; AI = Artificial Intelligence; AIPI = Artificial Intelligence Preparedness Index; CCA = Caucasus and Central Asia; EM = emerging market (includes both oil importers and oil exporters); EMMIs = emerging market and middle-income economies; GCC = Gulf Cooperation Council; LIC = low-income country; MENAP = Middle East, North Africa, Afghanistan, and Pakistan; UAE = United Arab Emirates.

C. AI Access

Disparities in access to advanced AI components—the availability of AI-specific technologies, data, and infrastructure—are deepening the technological divide across economies. AEs dominate the development and export of high-end chips, cloud-computing infrastructure, and frontier AI models, whereas most EMs and LICs face significant constraints in acquiring these inputs. Limited domestic manufacturing capacity, the high cost of specialized hardware, and dependence on a narrow set of foreign suppliers restrict these countries' ability to adopt and diffuse AI technologies at scale. As a result, the global distribution of AI capabilities is at the risk of becoming more unequal, with frontier AI innovations increasingly concentrated in economies that already benefit from strong digital ecosystems and well-resourced research environments.

Trade and technology restrictions, for example, because of geopolitical fragmentation may further reinforce this divide, particularly in regions characterized by uneven economic development, such as the Middle East. GCC countries—leveraging substantial financial resources, strategic partnerships, and proactive digital strategies—have achieved considerably greater access to cutting-edge AI components and platforms than many of their neighbors. For example, in early 2026, the United Arab Emirates and Qatar signed the non-binding US-led Pax Silica Declaration, a framework aimed at coordinating trusted supply chains for technologies underpinning the AI era, including semiconductors, critical minerals, advanced manufacturing, computing, logistics, and energy infrastructure.[4] Both Saudi Arabia and the United Arab Emirates have also secured access to advanced semiconductor technologies from the United States.

[4] The Pax Silica declaration, adopted in December 2025, commits participating countries to cooperate in building secure, resilient, and innovation-driven global technology supply chains. The initiative aims to strengthen economic security, reduce strategic dependencies, and promote trusted technological ecosystems among aligned partners. As of February 2026, nine countries had signed the declaration.

By contrast, countries outside the GCC, lacking comparable fiscal space and regulatory leverage, face a heightened risk of technological marginalization amid intensifying global competition and evolving export control frameworks. This divergence could entrench a dual-speed AI landscape within the region, whereby GCC economies advance more rapidly toward AI-enabled productivity gains, whereas other parts of the MENAP and CCA regions face rising barriers to technological catch-up. As shown quantitatively in Section 4, the macroeconomic impact of AI adoption depends not only on exposure and preparedness but also critically on access to the relevant AI technologies and infrastructure.

3. From Energy Powerhouse to AI Superhub

This section reviews the rapid expansion of AI-related investment across the MENAP and CCA regions. In several countries, these investments constitute an additional macroeconomic transmission channel through which AI can affect growth, productivity, and external balances. The section also examines the key drivers of demand for AI services—including the diffusion of AI technologies regionally and globally and the region's integration into global AI supply chains—and discusses how AI-related investment may support economic diversification.

A. The AI Investment Surge

Several MENAP and CCA economies have been undertaking large-scale strategic investments aimed at positioning themselves in the emerging AI ecosystem. These strategies build on structural advantages, including abundant low-cost energy, substantial financial resources, relatively flexible regulatory frameworks (Albous, Al-Jayyousi, and Stephens 2025),[5] and strategic geographic location. Together, these factors enable the development of large-scale, cost-efficient AI infrastructure and, in some cases, support ambitions to become AI-driven technology hubs.

However, the war in the Middle East that began in February 2026 has introduced substantial uncertainty, which may prompt some GCC economies to recalibrate diversification and investment priorities. It is also leading data center developers to re-evaluate plans for expanding AI and cloud data center capacity across GCC countries, as some data centers were damaged during the war.[6] These pressures are also reshaping the global AI landscape. Major hyperscalers are pausing or relocating data center investments previously earmarked for the Middle East.[7] Spillovers from regional tensions to global energy markets are raising the cost of powering AI workloads, while disruptions along key logistics routes are delaying deliveries of servers and specialized microchips. In response, global IT leaders are increasingly adopting multi-region cloud strategies to insulate their AI operations from localized physical risks, a shift that could erode advantages the Middle East and Central Asia economies had sought to consolidate.[8]

These risks arise in a context where venture capital investment in AI across the region has risen sharply since 2018 (Figure 4, panel 1). Although the annual AI venture capital investment in MENAP EMMIs (excluding the GCC)—about $100 million in 2024—remains small compared with the approximately $6 billion in AEs, cumulative growth in AI-related venture capital investment between 2018 and 2024 was four times faster in MENAP than in AEs, albeit from a lower base.

[5] In the GCC, AI governance has generally relied on "soft" regulatory approaches, with an emphasis on strategies and ethical guidelines rather than binding rules. This approach favors a more flexible and innovation-friendly model.

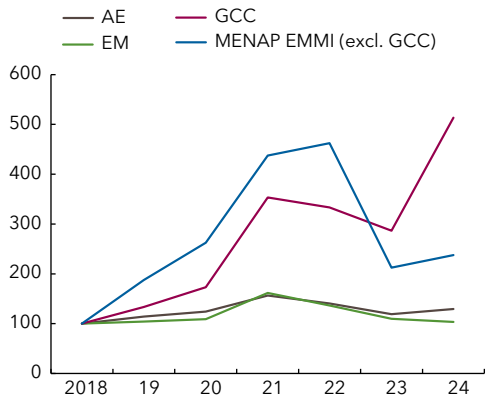
[6] Jeremy Hsu, "Drone strikes on data centers spook BigTech, halting Middle East projects," Ars Technica, April 29, 2026. https://arstechnica.com/ai/2026/04/data-center-developer-pauses-middle-east-projects-after-war-damage/

[7] Hyperscalers are the big technology companies that provide cloud computing and data storage on a global scale.

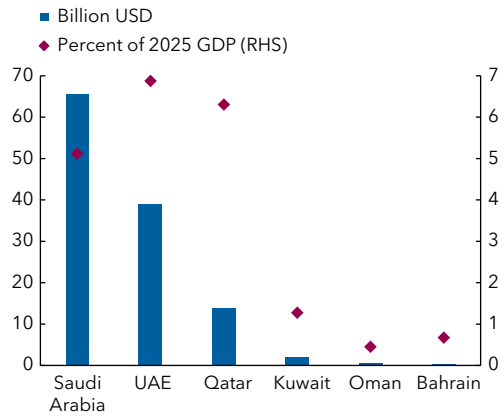
[8] Dug Nguyen, "AI and the Middle East War: Global Economic Impacts," AlQuinta, March, 2026. https://aiquinta.ai/insight/middle-east-war-affect-the-global-ai-industry/

Figure 4. AI-Related Investments in MENAP and CCA

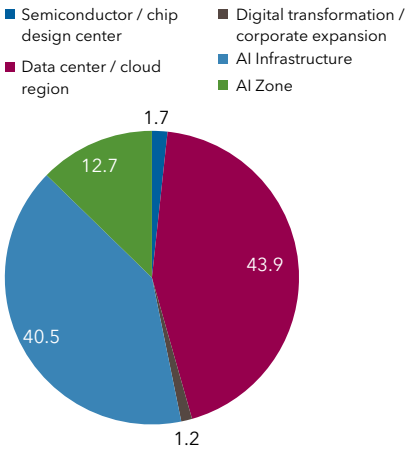
1. Venture Capital Investment Trends
(2018 = 100)



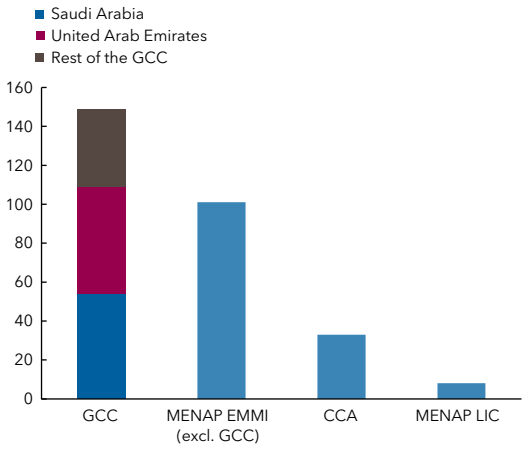
2. Announced Future Investments in AI
(As of end-2025)



3. AI-Related Investment Project Type
(Percent of total investments in the GCC)



4. Data Centers by Subregion
(Number)



Sources: DataCenterMap; Organization for Economic Co-operation and Development (OECD); and IMF staff calculations.

Note: For panel 2, figures reflect officially announced investment amounts. For panel 3, AI zones are designated physical or digital areas, intended to catalyze AI development, infrastructure build-out, and deployment. AI infrastructure projects refer to the specialized physical and digital backbone required to support AI development and deployment at scale, including components of data centers and cloud platforms not classified elsewhere, advanced computing hardware, and energy systems. AE = advanced economy; AI = artificial intelligence; CCA = Caucasus and Central Asia; EMMI = emerging market and middle-income economy; GCC = Gulf Cooperation Council; LIC = low-income country; MENAP = Middle East, North Africa, Afghanistan, and Pakistan; UAE = United Arab Emirates.

GCC countries, primarily through partnerships with global technology firms, are pursuing particularly ambitious strategies to establish themselves as global AI hubs (Box 5). In 2025, Qatar, Saudi Arabia, and the United Arab Emirates announced AI investment plans exceeding 5 percent of GDP—amounting to roughly $120 billion (Figure 4, panel 2)—placing the region prominently on the global AI investment landscape. Based on in-depth narrative analysis of project details disclosed in official investment announcements, these investments are primarily concentrated in data centers (44 percent) and other AI-related infrastructure (40 percent), with a smaller share allocated to the establishment of AI zones (13 percent) (Figure 4, panel 3). AI zones, such as those launched in Oman and Saudi Arabia, are designated physical or digital areas—often government-backed—designed to catalyze AI development, infrastructure expansion, and technology deployment. Although data centers are present in all MENAP and CCA subregions, they are heavily concentrated in GCC countries, followed by other MENAP EMMIs (Figure 4, panel 4).

In the CCA region, authorities are expanding AI-related infrastructure. Azerbaijan is advancing AI infrastructure via its 2025–2028 AI Strategy, including through operationalizing the South Caucasus' first TIER III data center and implementing a Government Cloud (G-Cloud) to centralize state IT systems and digital resources. Armenia is developing a large AI data center through a public–private partnership, whereas Kazakhstan is pursuing deeper integration into regional and global AI ecosystems, including through new data center investments and a memorandum to establish a regional AI compute hub (Box 4).

Although MENAP EMMIs (excluding the GCC) are less focused on becoming global AI hubs, they are nonetheless seeking to capture the growth potential of AI and digitization. Egypt, Jordan, and Morocco have adopted national strategies to promote AI and digitalization. Egypt aims to train 30,000 AI specialists through a partnership with Microsoft, and Jordan has launched the Jordan AI Venture Fund to support more than 60 AI start-ups (Box 3). Morocco has earmarked $1.2 billion for digital transformation and is developing sovereign data centers, including a large facility powered by renewable energy.

B. Chasing the AI Dividend: Where Will Demand Come From?

Global demand for AI services is expanding rapidly across regions, including MENAP and CCA. Rising AI diffusion, deepening digital ecosystems, the rapid emergence of AI-focused firms, and growing demand for specialized AI talent suggest that countries investing in AI infrastructure are well positioned to encounter growing interest and potential demand for the AI-related services they will provide.

The AI Diffusion Index shows a broad-based increase in adoption across regions in 2025, with particularly strong gains in AEs and the GCC (Microsoft AI Economy Institute 2026). MENAP EMMIs (excluding the GCC), the CCA, and MENAP LICs recorded more moderate—but positive—improvements, reflecting a broad rise in AI use during the second half of 2025 and indicating scope for further diffusion, along with associated demand for AI-related services (Misra and others 2025; Microsoft AI Economy Institute 2026) (Figure 5, panel 1).[9]

Additional indicators point to strengthening global demand. The number of newly established AI firms—defined as firms whose core business involves developing, applying, or commercializing AI technologies—increased sharply in 2024 across all regions, far exceeding the average pace of firm creation in the preceding decade (Figure 5, panel 2). This surge likely reflects expectations of sustained demand growth. These firms, in turn, generate additional demand for AI-enabling infrastructure.

At the same time, demand for AI-related skills is reshaping labor markets (Jaumotte and others 2026). LinkedIn job posting data indicate that hiring for AI-related skills rose markedly in 2024 relative to overall hiring in selected AEs, EMs, and GCC countries (Figure 5, panel 3). Together, these trends point to a strengthening ecosystem of AI-oriented firms and skilled labor, reinforcing growth prospects for economies supplying AI-related services.

A significant share of demand for AI services is likely to originate outside the MENAP and CCA regions.[10] GCC countries are positioning themselves to meet the needs of governments and firms—particularly those in AEs—operating under strict data privacy and localization laws. Initiatives such as "digital embassies" alongside broader efforts to integrate into global AI supply chains illustrate these strategies. Although physically located in the GCC, digital embassies would allow foreign governments to host sovereign digital infrastructure abroad while retaining full authority over data governance and operations. Bahrain laid early groundwork by adopting legislation in 2018 enabling foreign entities to procure cloud-computing services

[9] AI exposure captures the potential for productive AI adoption based on a country's employment structure, whereas AI diffusion refers to the share of the general population that has used a generative AI product during the reported period.

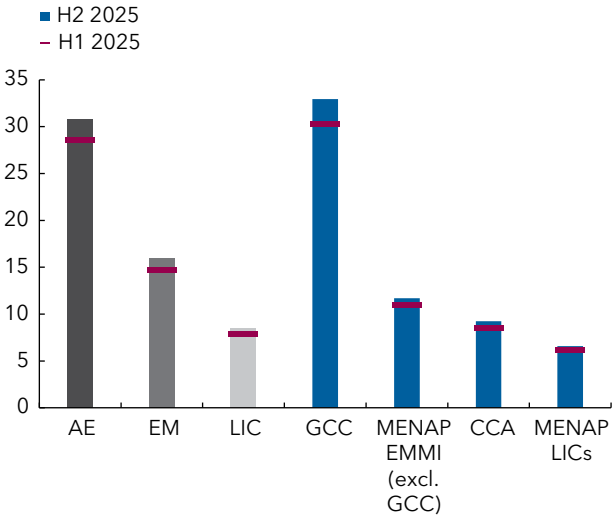
[10] AI infrastructure investments in the region serve a dual purpose: supporting domestic AI adoption while also facilitating the export of AI-related services.

while retaining domestic jurisdiction over their data, thereby supporting the development of technology-intensive sectors.[11] Saudi Arabia is reportedly exploring similar arrangements.[12]

Figure 5. AI Adoption in MENAP and CCA

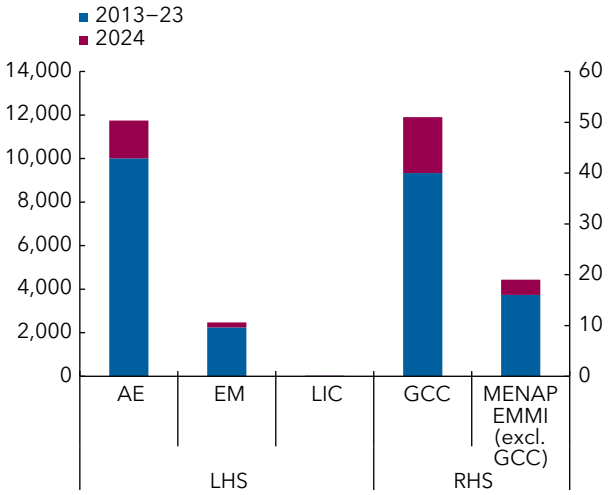
1. AI Diffusion

(Percent of total population, weighted by population per country)



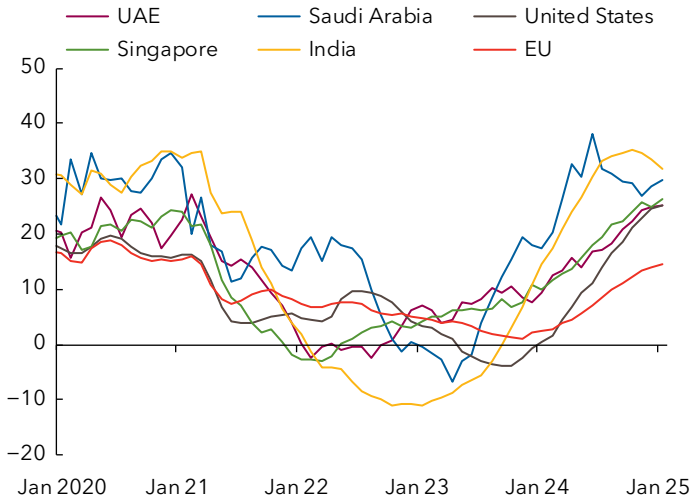
2. Number of AI Companies

(2013–24)



3. Relative AI Hiring Rate

(Percent, year-on-year ratio, 12-month moving average)



Sources: Brynjolfsson, Chandar, and Chen (2025); Microsoft AI Economy Institute (2026); OECD; and IMF staff calculations.

Note: Regional averages reflect country-level data availability. Panel 3 shows the growth rate of AI talent hiring relative to overall hiring in each country or region. For example, an AI hiring rate of 20 percent indicates that AI talent hiring has grown 20 percent faster than total hiring. AE = advanced economy; AI = artificial intelligence; CCA = Caucasus and Central Asia; EM = emerging market (including oil importers and exporters); EMMI = emerging market and middle-income economy; GCC = Gulf Cooperation Council; LIC = low-income country; MENAP = Middle East, North Africa, Afghanistan, and Pakistan; UAE = United Arab Emirates.

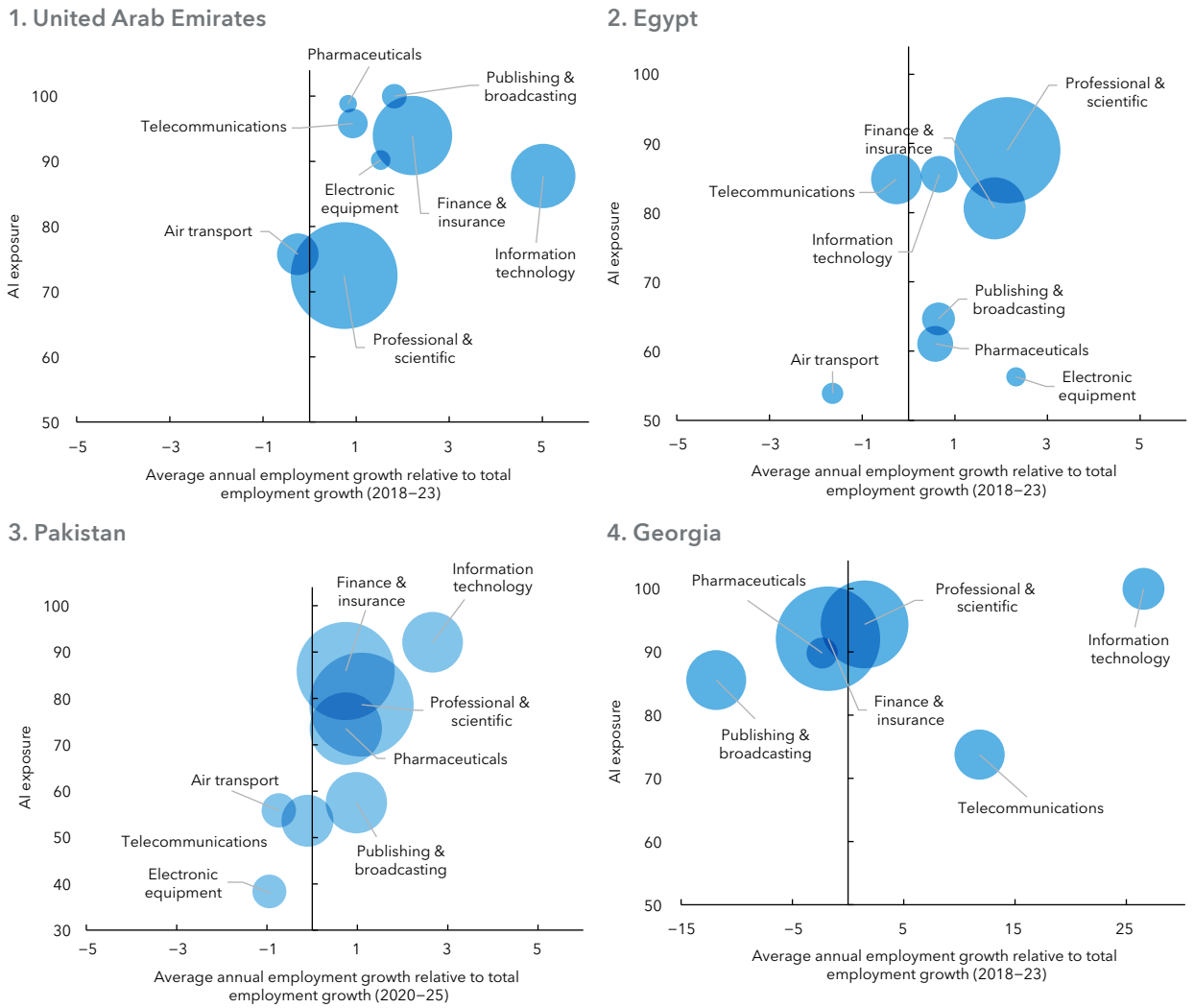
[11] As established under the *Law of Providing Cloud Computing Services to Foreign Parties*, Kingdom of Bahrain, accessed February 2026, https://bahrainbusinesslaws.com/laws/Law-of-Providing-Cloud-Computing-Services-to-Foreign-Parties.

[12] Tasmin Lockwood, "Saudi Arabia eyes data embassies amid sovereign AI push," *CNBC,* December 9, 2025. https://www.cnbc.com/2025/12/09/saudi-arabia-eyes-data-embassies-amid-sovereign-ai-push.html.

C. AI's Potential to Diversify the Economy

Economic diversification toward sectors with high AI exposure can raise the likelihood of AI adoption and amplify productivity gains across the economy. At the same time, AI itself can support diversification by enhancing productivity in nonresource sectors (Box 1) and enabling the expansion of emerging, high-growth AI-intensive industries (Figure 6). In the United Arab Emirates, for example, AI-intensive sectors—such as information technology and finance and insurance—combine high exposure to AI, relatively large employment shares, and strong recent growth, positioning them to contribute meaningfully to aggregate nonresource growth. Although other AI-intensive industries currently account for smaller employment shares, they show potential to expand and support economic diversification. Similar dynamics may emerge in the electronic equipment industry in Egypt and the United Arab Emirates and in the IT sector in Georgia and Pakistan.

Figure 6. AI-Intensive Sectors: AI Exposure and Growth
(Percent)



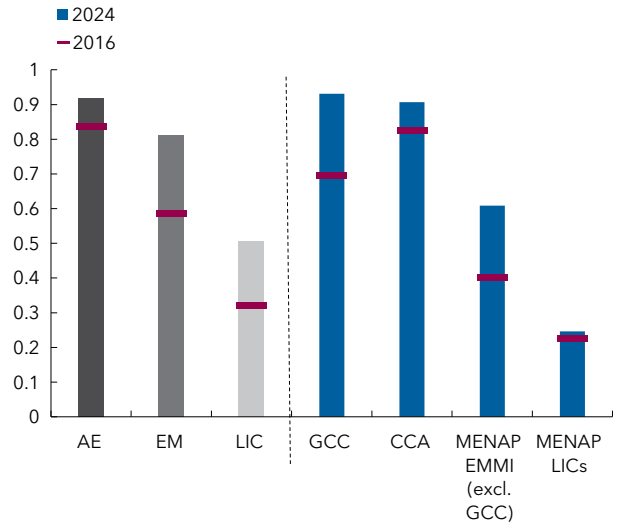
Sources: Cazzaniga and others (2024); International Labour Organization (ILO); and IMF staff calculations.

Note: AI exposure is measured as the share of employment in each industry that is highly exposed to AI. Bubble size reflects the sector's share of total employment in each country. AI = artificial intelligence.

Box 1. Examples of AI Adoption across Sectors

Artificial Intelligence (AI) adoption across the Middle East, North Africa, Afghanistan, and Pakistan (MENAP) and the Caucasus and Central Asia (CCA) economies is already reshaping activity in finance, energy, and public administration. Although implementation remains uneven, early applications illustrate how AI can improve efficiency, strengthen risk management, and enhance service delivery.

Box Figure 1.1. E-Government Development Index
(Index, higher = better, weighted by GDP)



Sources: UN E-Government Development Index; and IMF staff calculations.
Note: AE = advanced economy; CCA = Caucasus and Central Asia; EM = emerging market; EMMI = emerging market and middle-income economy; GCC = Gulf Cooperation Council; LIC = low-income country; MENAP = Middle East, North Africa, Afghanistan, and Pakistan.

Financial services. AI is transforming financial intermediation by improving operational efficiency, strengthening risk management, and expanding financial inclusion. Banks in the Gulf Cooperation Council (GCC) are deploying AI for personalized customer services (Saudi Arabia[1] and the United Arab Emirates[2]), fraud detection and transaction monitoring (Qatar[3] and Saudi Arabia[4]), and, in some cases, investment advisory functions (United Arab Emirates[5]). AI-driven credit-scoring tools are accelerating loan approvals and broadening access to finance, including for previously underserved or unbanked populations (Egypt[6,7]). Jordan and Egypt are using AI-enabled financial platforms to support start-up ecosystems and promote digital inclusion, contributing to the expansion of fintech services and digital payments.

Energy sector. AI is playing an increasingly important role in enhancing efficiency, resilience, and modernization in the energy sector. As a capital-intensive industry, energy production benefits from AI-driven optimization across hydrocarbon operations and supports the shift toward alternative energy sources. Across the value chain—particularly in the GCC—energy companies are using AI to optimize exploration and production processes, enable predictive maintenance to reduce downtime, improve operational efficiency and output, monitor and reduce emissions, enhance carbon-capture operations, strengthen cybersecurity for critical infrastructure, and improve grid management and renewable energy integration. For example, the State Oil Company of the Azerbaijan Republic (SOCAR) is developing "*Seismic AI*" for automated geological data analysis and implementing an AI-driven Smart Gas Network to optimize consumption, detect leaks, and reduce carbon emissions.

Public sector. Governments across the MENAP and CCA regions are increasingly adopting AI to enhance public service delivery. Broader digital readiness has improved over time: The UN E-Government Development Index shows gains across all country groups between 2016 and 2024 (Box Figure 1.1). The GCC now performs close to advanced-economy benchmarks, and the CCA and MENAP EMMIs (excluding the GCC) have made steady progress. MENAP low-income countries continue to lag, reflecting digital infrastructure and institutional constraints. AI applications in the public sector are expanding across the region. In Georgia, AI is used to detect anomalies in government payment requests. Qatar's "Ouqoul" platform applies AI to match graduates with employers. Azerbaijan has introduced AI-enhanced call centers to improve responses to tax and public service

inquiries and has integrated AI into its *unified E-Court Information System*, providing AI-powered search tools, digital filings, and analytics dashboards, significantly reducing case backlogs and improved transparency. In Saudi Arabia, AI supports health care delivery, education systems, and environmental monitoring. Armenia is piloting AI tools for tax compliance, risk management, and fraud detection.

[1] Al-Baity (2023).

[2] Ayush Kanodia, "Smarter, Faster Customer Support: AI Chatbots Transforming UAE Banking," WDCS Technology.ae, September 12, 2025. https://www.wdcstechnology.ae/smarter-faster-customer-support-ai-chatbots-transforming-uae-banking.

[3] "AI Fraud Detection in Banking: Saudi & UAE Focus," Makitsol.com, November 25, 2025. https://makitsol.com/ai-fraud-detection-in-banking-saudi-uae-focus/.

[4] Mark Bowen, "AlRajhi Bank partners with Mozn to harness AI in the fight against fraud," *Intelligent CIO Middle East,* October 20, 2025. https://www.intelligentcio.com/me/2025/10/20/alrajhi-bank-partners-with-mozn-to-harness-ai-in-the-fight-against-fraud/.

[5] "The Integration of Artificial Intelligence in Dubai Islamic Bank: A Technological Paradigm Shift," Cash-Platform.com, June 1, 2025. https://www.cash-platform.com/the-integration-of-artificial-intelligence-in-dubai-islamic-bank-a-technological-paradigm-shift/?utm.

[6] Ahmed Mosen, "How responsibly deploying AI credit scoring models can progress financial inclusion," *World Economic Forum,* October 2025. https://www.weforum.org/stories/2025/10/how-responsibly-deploying-ai-credit-scoring-models-can-progress-financial-inclusion/.

[7] Mayowa Kuyoro and Umad Bagus, with Anass Bensrhir and Ziyaad Bobat, "Leading, not lagging: Africa's gen AI opportunity," *McKinsey & Company*, May 12, 2025, https://www.mckinsey.com/capabilities/quantumblack/our-insights/leading-not-lagging-africas-gen-ai-opportunity.

4. The Macroeconomic Impact of AI

The section quantifies the potential medium-term macroeconomic gains from AI across the MENAP and CCA regions. It proceeds in three steps. First, it estimates growth gains from a plausible range of AI-induced productivity shocks. Second, it incorporates the additional impact of the AI-related investment surge in emerging AI hubs, under alternative assumptions about export growth. Third, it evaluates two policy-relevant scenarios of enhanced AI preparedness and restricted access to AI technologies.

A. Model

The analysis uses the Global Integrated Monetary and Fiscal Model (GIMF), a micro-founded, multiregion dynamic stochastic general equilibrium (DSGE) model originally developed by Kumhof and others (2010). The model incorporates heterogeneous households; financial, real, and nominal frictions; and detailed monetary policy and fiscal blocks.

International linkages are central to the model. Countries trade differentiated consumption, investment, and intermediate goods and are connected through financial markets and net foreign asset positions. These channels allow shocks in one region to spill over to others through movements in exchange rates, interest rates, and trade balances. Fiscal tools—including government spending and labor, capital, and consumption taxes—operate alongside monetary policy to determine inflation, output, and debt dynamics in a manner consistent with intertemporal budget constraints and overall external balance.

The model is calibrated to eight regional blocs. Six cover the MENAP and CCA regions: the United Arab Emirates, Saudi Arabia, the rest of the GCC, the CCA region, MENAP EMMIs (excluding the GCC) and MENAP LICs (see Annex Table 1.2 for details on country coverage). The remaining two blocs capture AEs and the rest of the world. In all scenarios, the AI productivity shock is global, affecting all blocs to an extent proportional to their current AI exposure, preparedness, and access.

The model extends the standard two-sector GIMF framework by incorporating global value chain linkages. In addition to the traditional tradable and non-tradable sectors, it adds a second tradable AI-intensive sector. This sector groups together industries with high AI exposure, including pharmaceuticals, computer and electronic equipment, and finance and insurance (Annex Table 1.3).[13]

Although the AI-intensive sector exhibits higher employment exposure—close to or above 80 percent across MENAP and the CCA—it represents a relatively modest share of GDP. It accounts for about 12 percent of GDP in the United Arab Emirates, 10 percent in Saudi Arabia, and 14 percent in the rest of the GCC. The CCA region and MENAP LICs have similar shares to the GCC (though smaller in absolute size), whereas in MENAP EMMIs (excluding the GCC), the share is about 8 percent (Table 1).[14] The inclusion of global value chain linkages allows the model to capture cross-country spillovers from AI shocks. This feature is key to analyzing the global impact of AI shocks and, specifically for this paper, the spillover of AI-related shocks in the GCC to the rest of MENAP and the CCA and feedback effects from developments in the rest of the world to the MENAP and CCA regions.

[13] In the model, the tradable sector captures the standard tradable industries related to agriculture, natural resources and commodities, and manufacturing. The non-tradable sector captures the remainder of the economy, primarily the retail trade and other domestic services.

[14] By comparison, AI exposure in the non-tradable sector ranges from about 25 percent in LICs to 46 percent in Saudi Arabia and the CCA. In the tradable sector, the exposure is lower, ranging from 6 percent in LICs to 38 percent in the rest of the GCC.

Table 1. GDP by Sector
(Percent of Regional/Country GDP)

	Saudi Arabia	United Arab Emirates	Rest of GCC	MENAP EMMI (excl. GCC)	MENAP LIC	CCA
AI-Intensive	9.8	12.1	14.1	8.4	13.5	11.7
Tradable	47.9	41.8	59.2	47.8	55.3	49.6
Non-tradable	42.3	46.1	26.6	43.8	31.2	38.7

Source: IMF staff calculations.
Note: AI = artificial intelligence; CCA = Caucasus and Central Asia; GCC = Gulf Cooperation Council; EMMI = emerging market and middle-income economy; LIC = low-income country; MENAP = Middle East, North Africa, Afghanistan, and Pakistan.

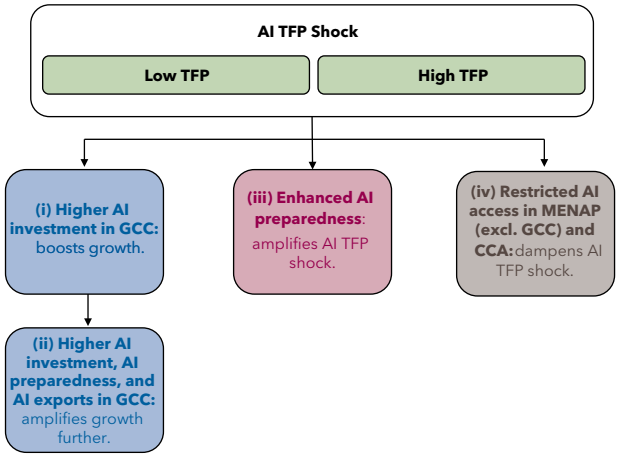
The dynamic effects of shocks in GIMF distinguish between the short term—when nominal and real rigidities tend to amplify the effects of shocks on aggregate demand—and the longer term. Over the medium to long term, the effects are driven by the permanent impact of productivity changes on the key inputs to production (capital and labor). Cerutti and others (2025) provide a detailed description of the model dynamics and the micro-foundations.

AI adoption is modeled as a region- and sector-specific total factor productivity (TFP) shock (Box 2). The starting point is an AI-induced TFP shock estimated for an AE that arises through AI adoption in sectors and occupations that currently have high AI exposure (Rockall, Tavares, and Pizzinelli 2025). Following Cerutti and others (2025), this benchmark is rescaled for each country based on its AI exposure, level of preparedness, and access. The resulting country-level shocks are aggregated to regional levels and introduced into the GIMF model to provide medium-term simulated effects on productivity, GDP, inflation, and external balances through capturing the linkages between the real, monetary, fiscal, and financial sectors as standard in a general equilibrium model. Given the uncertainty surrounding the magnitude of AI-driven TFP gains in AEs, the baseline effect is calibrated using both low- and high-TFP shocks to provide a plausible range of outcomes.

Beyond the baseline TFP effect of AI adoption, the analysis also incorporates the possible effects of the AI investment surge in emerging AI hubs—Saudi Arabia, the United Arab Emirates, and the rest of the GCC—in line with the announced investment plans presented in Section 3 (Figure 4, panel 2). Figure 7 summarizes the analytical framework: It begins with the baseline simulations, adds the effects of the AI investment boom in some countries, and then evaluates two key scenarios—enhanced preparedness and restricted AI access.

Despite providing valuable insights into AI's potential macroeconomic impacts, the model simulations require some simplifying assumptions. First, GIMF's treatment of labor markets is stylized, because it does not explicitly model unemployment dynamics or changes in labor force participation, which may matter when AI augments or automates specific occupations. Second, the assumed AI-related productivity shocks may not fully capture differences in the speed of AI diffusion across sectors, regions,

Figure 7. Outline of the Model Simulations: Baseline AI TFP Shock and Scenarios



Sources: IMF GIMF; and IMF staff calculations.
Note: AI = artificial intelligence; CCA = Caucasus and Central Asia; GCC = Gulf Cooperation Council; MENAP = Middle East, North Africa, Afghanistan, and Pakistan; TFP = total factor productivity.

and time frames, which could be impacted by various policies, such as infrastructure investments and fiscal policy. Third, by focusing on a few broad sectors, the model overlooks the wide range of differences within those sectors. Fourth, it does not consider the organizational and managerial complexities that often determine how AI is deployed. Finally, the paper does not address distributional effects within countries, further underscoring that these findings, while suggestive, should be viewed as scenario-based simulations rather than definitive forecasts.

Box 2. Calibration of the TFP Shock

Following Cerutti and others (2025), the model incorporates a calibrated sector-level total factor productivity (TFP) shock by region assumed to result from AI adoption. The magnitude of the TFP shock depends on each region's AI preparedness and access, as well as the sectoral exposure. Therefore, it differs at time *t* across regions, sectors, and scenarios. Specifically, a region-specific TFP shock is obtained by rescaling a benchmark TFP shock in country *b* proportionally to the product of region-specific (*i*) AI preparedness and exposure and then multiplying it by a region-specific access index. In the baseline simulations, we assume AI access equal to 1 for all groups (full access):

$$TFPshock_{i,t} = \frac{AI\ preparedness_i \times AI\ exposure_i}{AI\ preparedness_b \times x\ AI\ exposure_b} \times AI\ Access_i \times TFPshock_{b,t} \quad (1)$$

Then, the region-specific TFP shock is apportioned to each sector *j* according to the relative sectoral AI exposure within each region:

$$TFPshock_{i,j,t} = \frac{AI\ exposure_{i,j}}{AI\ exposure_i} \times TFPshock_{i,t} \quad (2)$$

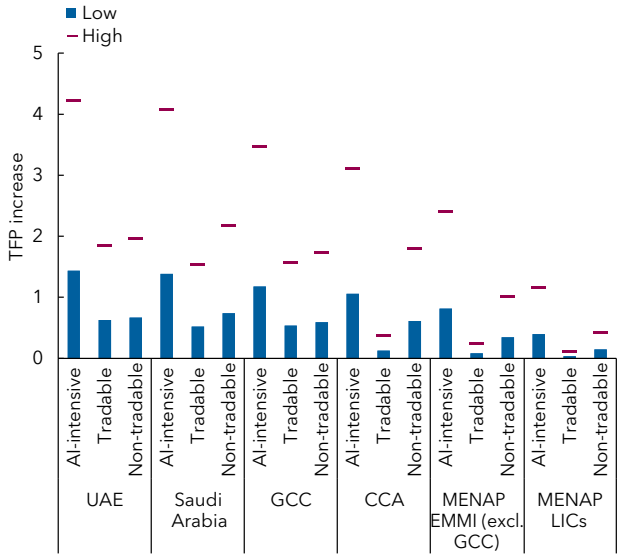
There remains much uncertainty around the magnitude of the TFP shock that could be driven by AI adoption. For example, two leading studies by Aghion and Bunel (2024) and Acemoglu (2025) arrive at notably different TFP gain estimates, given different underlying assumptions on AI exposure and the share of tasks that could profitably be replaced. Aghion and Bunel (2024) estimate a median annual TFP growth gain of 6.8 percent over 10 years for developed countries, assuming that 60 percent of tasks are exposed to AI and that 50 percent of those exposed tasks can be profitably replaced by AI. By contrast, Acemoglu (2025) estimates more conservatively that 4.6 percent of tasks in developed countries could profitably be replaced by AI, resulting in TFP gains of 0.7 percent over the next 10 years. The approach in this paper does not focus on worker displacement or augmentation, which would be important to study the within-country distributional effects, but discusses the uneven labor market exposure to AI adoption in Section 5.

Given the uncertainty around TFP shocks arising from AI adoption, the analysis uses both low and high benchmark TFP shocks to generate a range of possible impacts on output, investment, consumption, net exports, and other outcomes. These are based on simulations by Rockall, Tavares, and Pizzinelli (2025), who estimate a TFP increase of 1.3–3.4 percent over 10 years for the United Kingdom. Rescaling this range using AI preparedness and exposure (Equations 1 and 2) for the MENAP and CCA regional groups yields the TFP shock ranges shown in Box Figure 2.1. For the MENAP and CCA regions, the low-TFP scenario implies modest gains after 10 years (ranging from 0.1 percent in low-income countries to 0.8 percent for the United Arab Emirates), whereas the high-TFP scenario yields gains between 0.3 and 2.2 percent. Under this calibration, advanced economies experience TFP

increases of 1.2–3.4 percent over the same horizon. These estimates for MENAP and CCA (except for low-income countries) are larger than those for the United States as estimated by Acemoglu (2025), but they still represent a modest TFP effect when compared with Aghion and Bunel (2024).

Box Figure 2.1. Low and High AI-Related TFP Shocks

(Percent, increase after 10 years)



Source: IMF staff estimates.

Note: AI = artificial intelligence; CCA = Caucasus and Central Asia; GCC = Gulf Cooperation Council (in this figure, GCC excludes Saudi Arabia and the United Arab Emirates); EMMI = emerging market and middle-income economy (in this figure, EMMI excludes the GCC); LIC = low-income country; MENAP = Middle East, North Africa, Afghanistan, and Pakistan; TFP = total factor productivity; UAE = United Arab Emirates.

B. Baseline AI Productivity Shock

Emerging AI hubs, characterized by higher AI exposure and preparedness, are estimated to see the largest growth gains from the AI productivity shock. Depending on the magnitude of the TFP shock (low versus high), real GDP in the United Arab Emirates and Saudi Arabia are estimated to increase by about 0.2–0.5 and 0.1–0.4 percentage points per year on average, respectively. For the rest of the GCC, AI adoption could add about one-third of a percentage point to real GDP growth yearly.

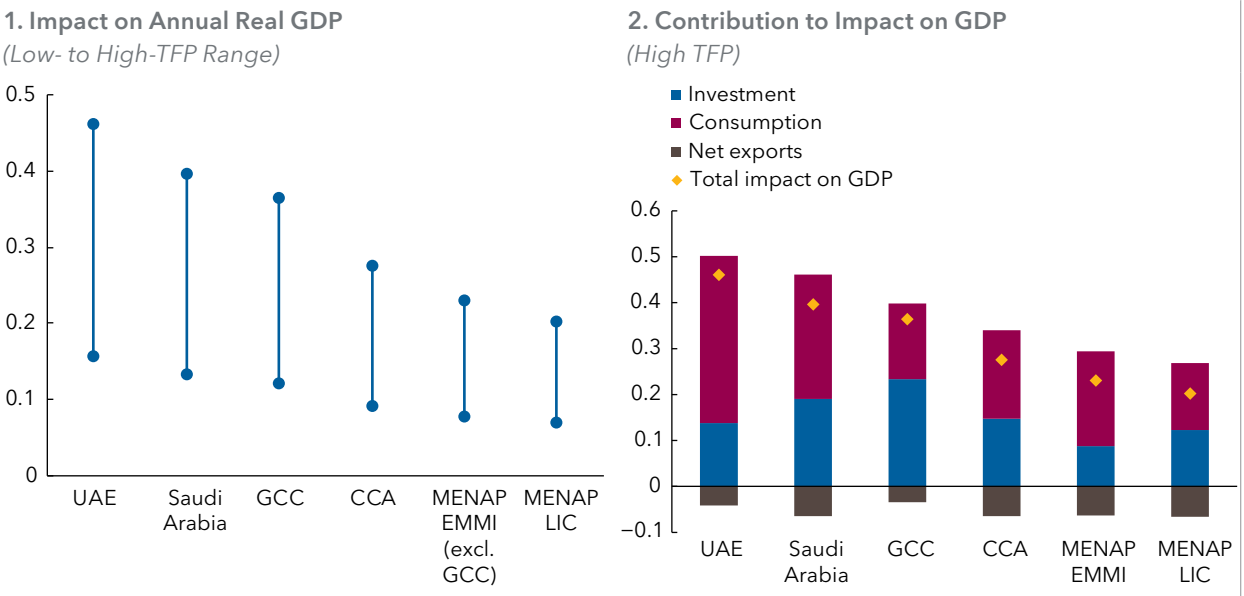
The growth impact varies significantly across country groups. Under the low- and high-TFP shock, annual growth gains range from 0.1 to about 0.25 percentage points in the CCA, from 0.08 to 0.23 percentage points for MENAP EMMIs (excluding the GCC), and from about 0.07 to 0.2 percentage points for MENAP LICs (Figure 8, panel 1). This implies that under the high-TFP assumption, growth gains in the United Arab Emirates and Saudi Arabia are roughly twice those of MENAP LICs and about 1.5 times those of the CCA region. These more modest gains reflect weaker AI preparedness and a sectoral employment structure that is less exposed to AI, particularly in LICs.

Across all country groups, the growth impact is driven primarily by consumption and investment (Figure 8, panel 2, high-TFP case). Although stronger global growth boosts exports from MENAP and CCA,

consumption and investment are relatively import intensive. As a result, imports are estimated to grow relatively faster than exports on average over the next decade, leading to a small negative contribution of net exports to GDP growth.

Figure 8. AI Productivity Shock Results

(Percent deviation from no shock scenario, 10-year average annual impact)



Sources: IMF GIMF; and IMF staff calculations.

Note: AI = artificial intelligence; CCA = Caucasus and Central Asia; GCC = Gulf Cooperation Council (in both panels, GCC excludes Saudi Arabia and the United Arab Emirates); EMMI = emerging market and middle-income economy (in this figure, EMMI excludes the GCC); LIC = low-income country; MENAP = Middle East, North Africa, Afghanistan, and Pakistan; TFP = total factor productivity; UAE = United Arab Emirates.

These simulated growth estimates incorporate the effect on other macroeconomic variables (such as inflation, imports, and exports) through the general equilibrium linkages in the model. The strong demand-side boost, driven by the positive lifetime income effects of anticipated higher future productivity, raises inflationary pressures in the short term.[15] Inflation peaks about three years after the shock before moderating and returning to its no-shock level (Figure 9, panel 1).[16] The impact is larger in Saudi Arabia and the rest of the GCC, where inflation is projected to rise by nearly 0.55 percentage point after three years, compared with about 0.45 percentage point in the other country blocs. The model allows for monetary policy adjustment, assuming central banks would tighten policy rates in the short term in response to higher inflation. This response applies to the CCA region and MENAP EMMIs and LICs. In the GCC, however, policy rates move in line with US rates because of pegged exchange rate regimes.[17] Over the longer term, inflation returns to the no-shock level as tighter monetary conditions and higher aggregate supply (from higher productivity and a larger capital stock) offset the initial demand boost.

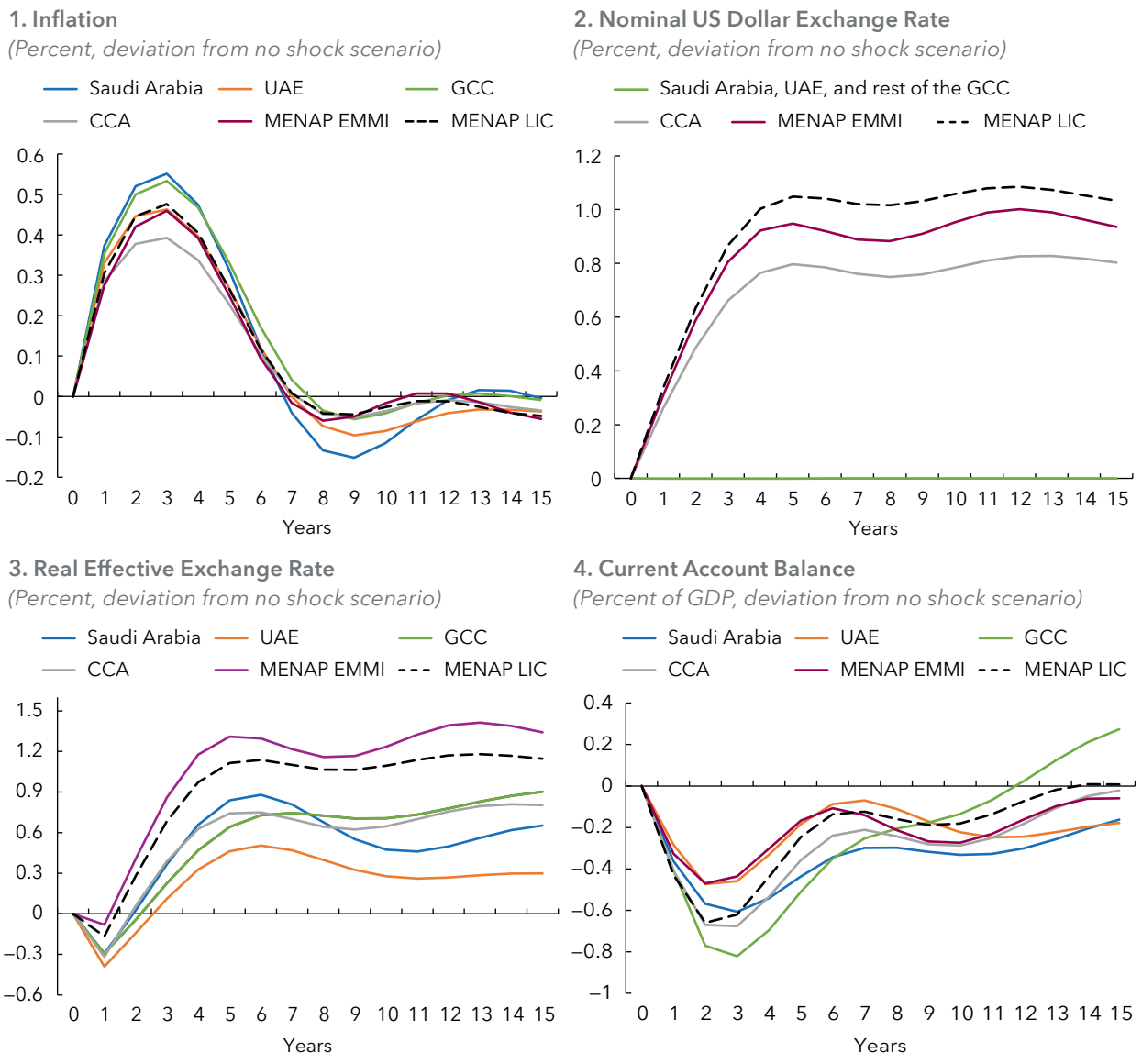
[15] Models assuming cost-reducing effects of AI and faster productivity pass-through could yield more muted or even disinflationary short-term outcomes.

[16] This finding relates to Aldasoro and others (2024), who analyze the impact of AI on output, employment, and inflation in a closed economy, multisector general equilibrium model. In their framework, inflation depends crucially on households' and firms' expectations. If future productivity gains are not anticipated, AI adoption may have a short-term disinflationary effect. Moreover, in models that incorporate unemployment, initial inflationary pressures may be further muted to the extent that AI adoption leads to some job displacement.

[17] In the United States, inflationary pressures would prompt an increase in the policy rate. Given the pegged exchange rate regimes in the GCC, monetary policy therefore moves in line with US policy, which may be imperfectly aligned with local macroeconomic conditions.

In MENAP EMMIs (excluding the GCC) and LICs—and to a lesser extent, the CCA region—where nominal exchange rates are flexible, the AI productivity shock results in sustained nominal and real appreciation (Figure 9, panels 2 and 3). As discussed in Cerutti and others (2025), the large increase in productivity in the non-tradable sector of the United States and other AEs relative to other countries leads to a real depreciation of their currencies relative to others (Inverse Balassa-Samuelson effect). Given that the difference in the productivity shock between the US and MENAP LICs and EMMIs (excluding the GCC) is the widest, the appreciation of the real effective exchange rate in these two blocs is larger. The appreciation makes imports relatively cheaper and exports relatively less competitive, leading to a temporary deterioration in current account balances across regions, which peaks at two to three years after the productivity shock before gradually returning toward the no-shock level (Figure 9, panel 4).

Figure 9. Baseline Impact on Inflation, Nominal Exchange Rate, Real Effective Exchange Rate, and Current Account (High TFP)



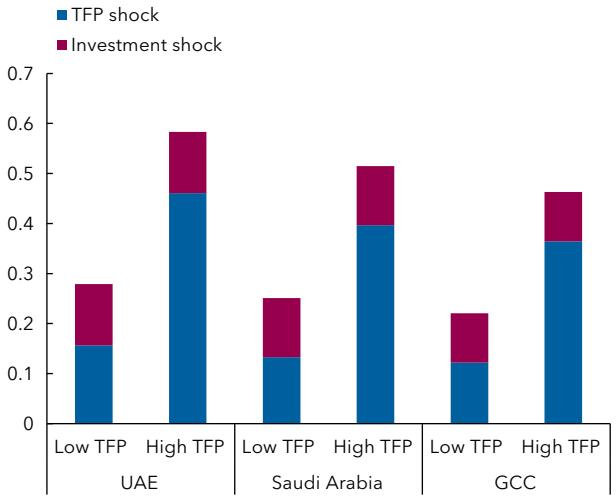
Sources: IMF staff calculations.

Note: CCA = Caucasus and Central Asia; GCC = Gulf Cooperation Council (in all panels, GCC excludes Saudi Arabia and the United Arab Emirates); EMMI = emerging market and middle-income economy (in this figure, EMMI excludes the GCC); LIC = low-income country; MENAP = Middle East, North Africa, Afghanistan, and Pakistan; TFP = total factor productivity; UAE = United Arab Emirates.

C. The GCC's AI Investment Wave

To understand the potential macroeconomic impact of the AI investment boom in the emerging AI hubs—Saudi Arabia, the United Arab Emirates, and the rest of the GCC—the analysis simulates an AI-related investment shock consistent with announced investment plans (Figure 4, panel 2). In addition to increasing the capital stock in the AI-intensive sector, the investment surge is assumed to raise AI preparedness in these economies to advanced-economy levels, reflecting improvements in digital infrastructure, skills, and institutional frameworks. Thus, these results represent an upper-bound scenario dependent on successful reforms.

Figure 10. Investment Wave: Impact on Real GDP
(Percentage point deviation from no shock scenario, 10-year average annual impact)



Sources: IMF GIMF; and IMF staff calculations.
Note: GCC = Gulf Cooperation Council (excluding Saudi Arabia and the United Arab Emirates); TFP = total factor productivity; UAE = United Arab Emirates.

Under current trade patterns, the AI investment wave in the GCC is projected to deliver meaningful but modest additional growth gains. Real GDP growth increases by about 0.1 percentage point per year over the next decade (Figure 10). Under the high-TFP scenario, total growth gains rise to nearly 0.6 percentage point annually for the United Arab Emirates and 0.5 percentage point in Saudi Arabia and the rest of the GCC.[18]

Although this investment shock scenario results in higher AI-related exports from the AI hubs, it takes on a conservative assumption on the growth of AI-related exports in that it assumes no change to current AI-export growth, contrary to the intended outcome of such investment strategies. In principle, GCC economies could expand exports of AI services more rapidly as they develop comparative advantages in providing more affordable AI-related services, supported by lower energy costs and large-scale AI infrastructure.[19] Stronger AI adoption abroad, improvements in regulatory alignment, and greater global reliance on offshore data centers are other factors that could support the GCC's faster AI-related export growth.

To capture this possibility, the analysis considers an investment scenario in which GCC countries secure a higher global market share in AI-intensive exports.[20] The results suggest sizable additional gains (Figure 11, panel 1). For the rest of the GCC, the growth impact of the investment shock with a higher export market share is roughly double that under the baseline export assumption (without increasing global market share). For Saudi Arabia, gains are about 90 percent higher, and for the United Arab Emirates, they are about two-thirds higher. Because the United Arab Emirates already has a higher share of AI-intensive exports, the gain from expanding market share is slightly smaller than for the rest of the GCC, where current AI-related exports are more limited. In addition, export growth is projected to rise by close to 0.4 percentage point annually over 10 years across the three AI hubs—about 0.3 percentage point more than in the investment shock scenario without the increase in global market share (Figure 11, panel 2).

[18] Although the discussion of the simulations focuses on the growth impact, similar general equilibrium dynamics as noted in relation to Figure 9 are also at play here. Notably, the investment boom along with the TFP impact is slightly less inflationary because investment also contributes to expanding supply, partially offsetting the demand-side inflationary pressures.

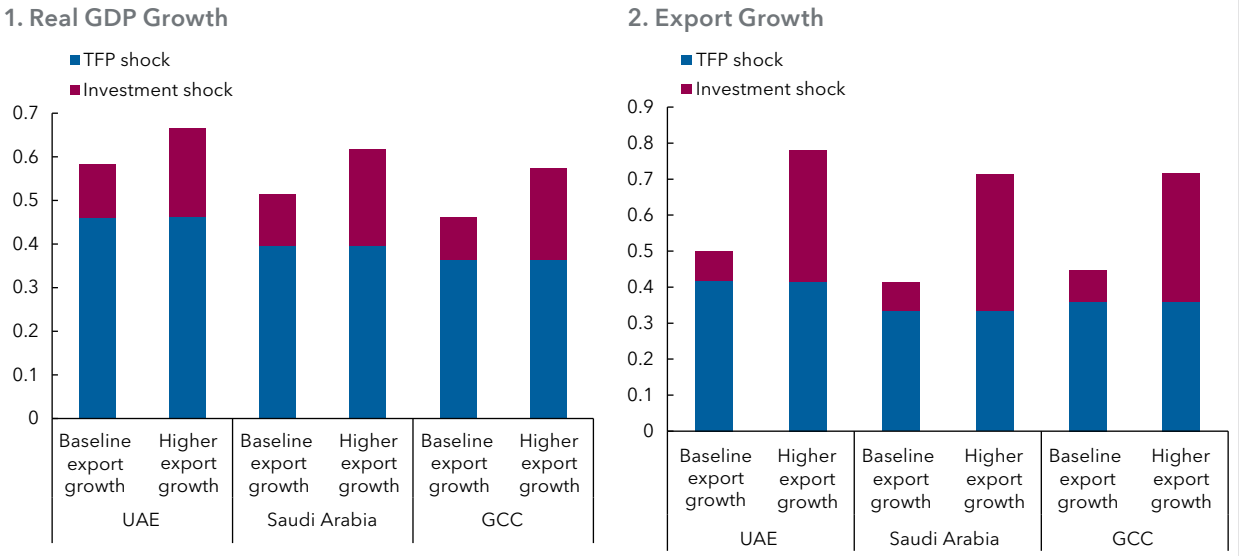
[19] Although energy costs are expected to rise as energy demand increases with more AI data centers in many regions of the world (see IMF, 2025 for estimates of expected AI-induced changes in electricity prices in the US, Europe, and China), the GCC economies are already starting at a lower-cost point and could maintain this relative advantage over time.

[20] The assumption in the model is that global export shares of AI-intensive goods increase by around 25 percent on average.

These results also highlight strategic complementarities between the GCC AI hubs and the broader global economy. In particular, greater progress in AI preparedness in lower-income MENAP and CCA economies and ensuring mutually convenient regulatory agreements with AEs could boost global demand for AI services and help GCC countries amplify the macroeconomic gains from their investment. However, should these planned AI investments not fully materialize or demand for AI services remain subdued in the rest of MENAP, the CCA, and other regions of the world, it would dampen the potential growth impact of the AI investment surge in the GCC.

Figure 11. Investment Wave with Higher Export Growth

(Percentage point deviation from no shock scenario, 10-year average annual impact)



Sources: IMF GIMF; and IMF staff calculations.

Note: In panel 1, the baseline export growth bars correspond to the high-TFP results shown in Figure 10 and are reproduced here for comparison. GCC = Gulf Cooperation Council (excluding Saudi Arabia and the United Arab Emirates); TFP = total factor productivity; UAE = United Arab Emirates.

D. Alternative Scenarios on the AI-Driven Productivity Boost

This section examines two alternative scenarios relating to how well AI is used (enhanced AI preparedness) and whether AI can be used (restricted access to AI technologies) in the MENAP and CCA regions.[21]

Given that several MENAP and CCA economies are not planning large AI-specific investments, this scenario illustrates the positive role of improvements in AI preparedness, for example, through successful reforms or implementation of a national AI strategy in countries that have adopted one. Under the "enhanced AI-preparedness" scenario, AI preparedness in the GCC and the CCA is raised to advanced-economy levels, whereas preparedness in MENAP EMMIs (excluding the GCC) and LICs is raised to the level of EMs. This increase proportionally raises the TFP shock across all subregions.

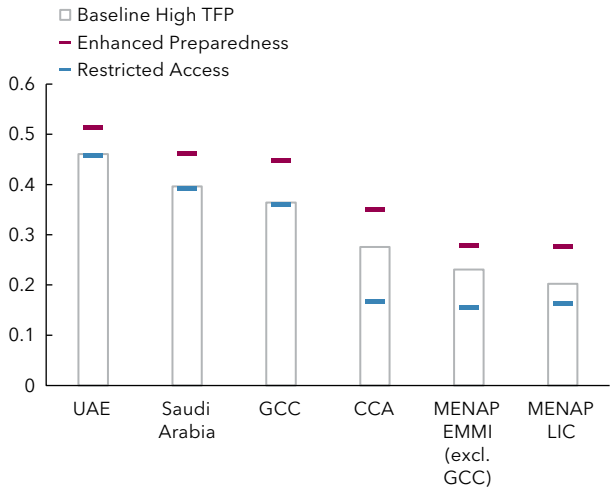
Simulation results (Figure 12) show that if the United Arab Emirates, Saudi Arabia, the rest of the GCC, and the CCA region raise AI preparedness to advanced-economy levels, annual real GDP growth could increase by an additional 0.05 percentage point in the United Arab Emirates and up to 0.1 percentage point in the rest of the GCC over a 10-year horizon. Similarly, if MENAP EMMIs (excluding the GCC) and LICs raise preparedness to emerging-market levels, annual growth could increase by about 0.05 and 0.1 percentage points, respectively. Gains are larger where initial preparedness gaps are wider.

[21] For brevity, the analysis focuses on the high-TFP case in all scenarios, noting that the conclusions remain consistent with the low-TFP case, albeit at a smaller scale.

By contrast, in a global environment characterized by heightened trade and geopolitical tensions, EMs and LICs face uncertainty regarding access to the required AI-related hardware and services, including advanced processors, frontier models, and data center capacity. If access to these inputs were constrained—whether because of export restrictions or geopolitical considerations—their ability to adopt AI across productive sectors would be limited, reducing potential productivity gains. In a pessimistic restricted-access scenario, MENAP EMMIs (excluding the GCC), LICs, the CCA region, and the rest of the world (excluding the GCC and AEs) face constraints on the availability of AI-specific technologies, leading to a significant reduction in AI-induced TFP growth for these groups.[22]

Figure 12. Enhanced Preparedness and Restricted Access: Impact on Real GDP

(Percentage point deviation from no shock scenario, 10-year average annual impact)



Sources: IMF GIMF; and IMF staff calculations.

Note: CCA = Caucasus and Central Asia; GCC = Gulf Cooperation Council, GCC excludes UAE and Saudi Arabia; EMMI = emerging market and middle-income economy; MENAP = Middle East, North Africa, Afghanistan, and Pakistan; LIC = low-income country; TFP = total factor productivity; UAE = United Arab Emirates.

If AI access is restricted in the MENAP (excluding the GCC) and CCA regions, annual real GDP growth would be lower by between 0.04 percentage point (LICs) and 0.1 percentage point (CCA) over 10 years relative to the baseline (Figure 12). In proportional terms, this reduces baseline gains by about 20 percent in LICs and 40 percent in the CCA region. Such constraints would widen the divergence that emerges from the baseline shock between the AI hubs (the GCC) and the rest of the MENAP and CCA region. Given that the TFP boost under the baseline for smaller economies such as LICs derives largely from the AI TFP boost in the rest of the world and the concomitant global demand for LICs' exports, the restricted-access scenario has a smaller dampening effect on real output.

[22] In the restricted-access scenario, the AI access variable (see Box 1) is assumed to remain equal to one in AEs and the GCC, whereas it is reduced to 0.25 for MENAP EMMIs (excluding the GCC), the CCA, MENAP LICs, and the rest of the world.

5. Risks and Policy Challenges

AI not only holds considerable potential to boost productivity and long-term economic prospects but also introduces important risks and policy challenges. Although not fully captured in the model simulations, in the near term, large and increasingly leveraged AI-related investment cycles may heighten financial-market vulnerabilities—raising the prospect of valuation corrections, credit-market stress, and cross-border spillovers. These risks are particularly relevant where banks hold substantial amounts of government securities, intensifying the existing sovereign–bank–real economy nexus. In the GCC, banks' holdings of government securities remain higher than in many EMs and developing economies (AlHassan and others 2022). The war in the Middle East has further amplified these risks for GCC economies, with investor sentiment likely to depend critically on the restoration of sustained regional stability and the preservation of the region's safe-haven status.

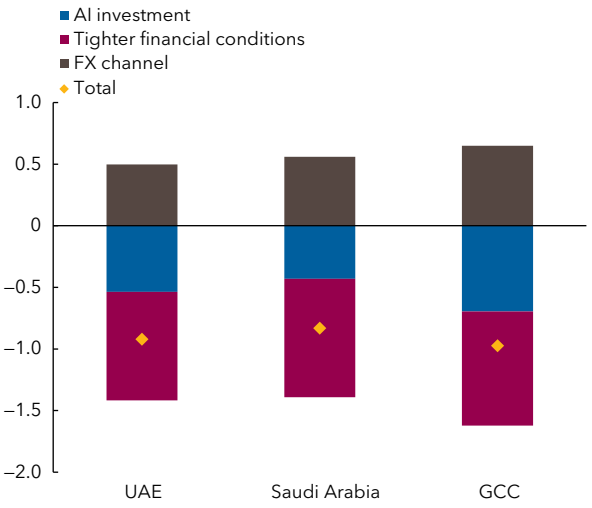
Over the medium term, broader AI adoption may intensify labor market pressures. Automation and skill-biased technological change could displace certain categories of workers, widen skill mismatches, and increase inequality. Although these developments may not materially affect the AI-related productivity estimates discussed in Section 4A, they may influence the dynamics of the growth effects and may also weaken tax bases and raise demand for social protection, creating fiscal pressures. Taken together, these short- and medium-term risks underscore the need for proactive macrofinancial oversight and forward-looking labor market and fiscal policies to ensure that AI adoption strengthens economic resilience and inclusion rather than amplifying existing structural fragilities.

A. Financial Market Risks

Rapid expansion of AI-related investment poses significant financial stability risks. If adoption across sectors or trading partners proves weaker than expected, large-scale AI infrastructure could become underutilized, leading to stranded digital assets and lower returns. These risks are compounded by sizable sovereign and corporate exposures: governments, public investment funds, and major technology firms often co-finance AI infrastructure, so an AI-related downturn could generate substantial macrofinancial spillovers. This concern is especially relevant for the GCC, where sovereign exposure to AI-linked investments has increased.

Figure 13. Impact of AI-Led Tech Downturn on Output

(Percent deviation from no shock scenario, cumulative over two years*)*



Sources: IMF GIMF; and IMF staff calculations.

Note: AI = artificial intelligence; GCC = Gulf Cooperation Council (excluding Saudi Arabia and the United Arab Emirates); UAE = United Arab Emirates.

To illustrate these risks, the analysis considers a downside scenario for the GCC region triggered by a reappraisal of expected productivity gains from AI, leading to a decline in investment in the technology sector of about 0.5–1.0 percent in the GCC countries. The scenario also includes a layer of lower asset prices and tighter financial conditions. Global equity prices outside the US fall by 15 percent, triggering a broader tightening of global financial conditions. Corporate risk premiums

and term premiums of sovereign bonds both temporarily rise by 75–100 basis points in EMs (IMF 2026). The adverse effects of these on the GDP would be partially offset by a depreciation of the US dollar, which would translate into monetary easing in economies with exchange rate pegs to the dollar. As shown in Figure 13, the cumulative negative impact on GDP amounts to about 1 percent over two years. This effect may compound with more rapid obsolescence of digital infrastructure if adoption is more subdued than predicted.

Financing structures further heighten vulnerabilities. Hyperscale data center projects are often highly leveraged, increasing the potential for credit-market stress, including debt-servicing difficulties, rating downgrades, and spillovers to bank and nonbank financial institutions with significant exposure to technology-linked assets. Sudden reversals in cross-border capital flows could amplify liquidity pressures in economies pursuing AI-intensive growth strategies. Large AI investment cycles may also shift the natural rate of interest and affect monetary policy transmission (Melina and Villa 2025), raising the risk of policy miscalibration.

Taken together, these dynamics underscore the need for robust prudential oversight, strengthening institutional capacity, closely monitoring the pace and scale of AI adoption, and systematically integrating AI-specific scenarios into macrofinancial policy frameworks (Shabsigh and Boukherouaa 2023; Breeden, 2024). Enhancing AI-related data collection is also critical to better detect vulnerabilities, support regulatory compliance, and improve supervisory data analytics (Financial Stability Board 2025).

B. Labor Market Risks

The economic impact of AI adoption across MENAP and CCA economies will vary by occupation and demographic group, depending on each occupation's exposure to AI—how closely AI capabilities overlap with required tasks—and its complementarity with AI. High exposure combined with low complementarity implies a greater risk of task automation, reduced labor demand, and downward pressure on employment and wages. By contrast, occupations characterized by both high exposure and high complementarity—such as executives, engineers, scientists, and professional services—are more likely to experience productivity and wage gains, provided workers possess the skills needed to use AI tools effectively. Low-exposure occupations tend to include manual and craft-based roles such as plumbing, electrical work, construction, and agriculture.[23]

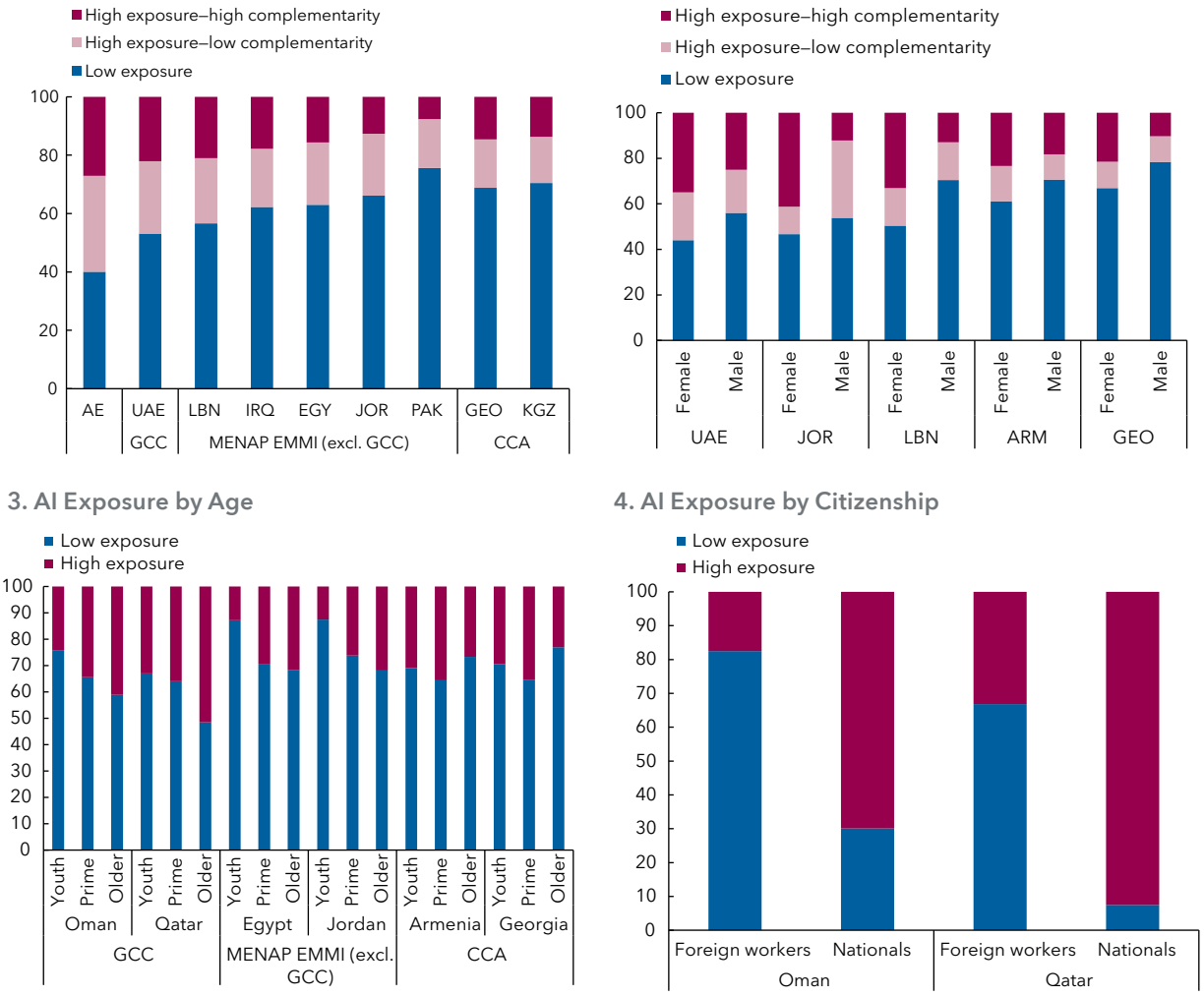
AI-exposed jobs account for 30–50 percent of employment across the MENAP and CCA economies, compared with about 60 percent in AEs, where employment is focused in cognitive-intensive roles (Figure 14, panel 1). Across the MENAP and CCA regions, women tend to face greater exposure but are also relatively well positioned to benefit from AI because of high complementarity, consistent with findings for other regions (Cazzaniga and others 2024) (Figure 14, panel 2). Younger workers are generally less exposed—except in the CCA (Figure 14, panel 3)—but evidence from other countries suggests they are more likely to be employed in occupations with lower complementarity to AI, a distinction this paper cannot assess fully because of limited granular data (Cazzaniga and others 2024; Jaumotte and others 2026). Emerging evidence from the United States indicates that generative AI adoption has reduced entry-level hiring, particularly in occupations where tasks are more easily automated than complemented by AI (Brynjolfsson, Chandar, and Chen 2025). As AI adoption progresses, similar dynamics could emerge in MENAP and CCA countries, especially in economies already facing high youth unemployment. In the GCC, nationals are disproportionately employed in higher-exposure occupations relative to foreign workers but are also likely to be employed in high complementary occupations (Figure 14, panel 4).

The broader diffusion of AI creates interconnected labor market and fiscal risks that demand proactive policy intervention. Automation threatens substantial job displacement in routine occupations, potentially straining unemployment systems. At the same time, rapid technological change is expected to generate

[23] High informality in the MENAP region may have implications on the macroeconomic impact of AI in the economy. The lack of social security within the informal sector can widen economic inequality, but AI can also present an opportunity to integrate informal workers into broader economic systems (Mpedi and Marwala 2025).

skill mismatches, because displaced workers may lack the competencies required for newly emerging roles, underscoring the need for large-scale and timely investments in reskilling and human capital. These dynamics risk widening income inequality, with highskill workers capturing productivity gains whereas lowskill workers face stagnant wages or prolonged unemployment, thereby increasing demands on social protection systems. Large-scale reallocation of labor across sectors and regions also entails significant adjustment costs, particularly in economies with limited labor mobility or weaker labor market institutions. A potential erosion of labor income tax bases—although offset in part by broader efficiency gains from AI (Brollo and others 2024)—combined with higher expenditures on unemployment benefits, retraining, and social support could intensify fiscal pressures. Uneven distributional effects may also undermine social cohesion and public trust. These risks underscore the importance of transparent and equitable policy frameworks to ensure that the gains from AI adoption are broadly shared.

Figure 14. AI Exposure and Complementarity
(Share of employment)



Sources: Cazzaniga and others (2024); ILO; and IMF staff calculations.

Note: Age groups are defined as follows: youth (15–24 years), prime (25–64 years), and older workers (55–64 years). AE = advanced economy; AI = artificial intelligence; CCA = Caucasus and Central Asia; EMMI = emerging market and middle-income economy; GCC = Gulf Cooperation Council; MENAP = Middle East, North Africa, Afghanistan, and Pakistan.

6. Conclusion

AI adoption across the MENAP and CCA regions, combined with AI-specific investment surges in the GCC, could raise real GDP growth by between 0.1- 0.6 percentage points per year across countries and scenarios. Growth gains in the GCC economies could be roughly 1.5 times larger than in the CCA region and about twice those in MENAP LICs.

Although structural differences across MENAP and CCA regions are likely to persist, policy reforms can help narrow the regional AI divide. The enhanced AI-preparedness scenario suggests that reforms could add up to 0.1 percentage point to annual real GDP growth in countries that strengthen their AI readiness. In economies with strong foundational AI preparedness (such as digital infrastructure and skills), emphasis should be placed on strengthening digital innovation capacity and adapting legal frameworks to govern and foster AI advances. Where foundational preparedness is limited, priority should be given to expanding digital infrastructure and investing in human capital to reap early gains from AI and build the foundations for more advanced AI adoption over time.

Improving access to AI technologies—including hardware, software, and, critically, relevant data—could further reduce the gap between AI front-runners and lagging economies. However, geopolitical and national security considerations may influence technology-sharing practices, with constraints often weighing most heavily on EMs and LICs that have fewer alternative channels for AI-intensive investment and innovation. The recent war in the Middle East could further complicate this landscape by increasing risk premia, disrupting investment plans, and possibly requiring a more security-driven approach to technology partnerships and data governance. For MENAP and CCA economies, preserving open, reliable, and rules-based channels for technology diffusion will therefore be essential to prevent geopolitical fragmentation from translating into a deeper AI divide.

AI investment offers a strategic channel to accelerate diversification and contain potential longer-term economic scarring from the war. This war has sharpened the urgency of broadening growth drivers in the GCC, a region where fiscal and external positions remain closely tied to oil revenues. Sustained AI investment can support this transition by anchoring high-productivity tradable activities outside the hydrocarbon sector, expanding export streams less correlated with oil-price cycles, and reinforcing medium-term policy credibility.

Despite the risks, GCC economies are expected to capture disproportionately larger gains, reflecting sectoral and employment composition more conducive to AI adoption, stronger AI readiness, superior access to advanced technologies, and plans to expand AI-specific digital infrastructure. These advantages are underpinned by abundant low-cost energy, substantial sovereign wealth resources supporting large-scale AI infrastructure, industrial strategies, and regulatory frameworks that facilitate agile implementation of AI-friendly regulation (World Economic Forum 2026; Albous, Al-Jayyousi, and Stephens 2025). Realizing these gains, however, will depend critically on sustained external demand for AI services, both regionally and globally. Strengthening AI preparedness in other MENAP and CCA economies—and securing sound regulatory agreements with AEs—could support higher AI exports, particularly as AEs account for a large share of global demand for AI services. Foreign investor sentiment will be pivotal in determining whether external demand and cross-border investment continue to support the region's AI ambitions, particularly in the GCC. The war in the Middle East may have weighed on investor confidence and perceptions of regional stability. A prolonged period of uncertainty could delay investment decisions, raise financing costs, and weaken the credibility of the region's diversification strategy. Conversely, a durable return to stability would help preserve the GCC's safe-haven appeal and its ability to attract the capital and technology partnerships needed to scale AI infrastructure.

For the MENAP (excluding the GCC) and CCA regions, lower expected growth gains from AI compared with the GCC reflect not only weaker preparedness but also lower sectoral exposure, given current economic structures. Nonetheless, these economies may still benefit from targeted adoption of more accessible and cost-effective AI applications—particularly in health, education, and public service delivery—often referred to as "small AI" (World Bank 2025).[24] Over time, AI adoption may also support economic diversification by shifting their economic and employment structure toward sectors with greater AI exposure, increasing the scope for AI productivity gains.

At the same time, AI adoption is not a substitute for broader structural reforms. Governance reforms, improvements in labor and product market flexibility, and greater trade openness are crucial to support dynamic, private sector–led growth. These reforms would not only raise potential growth directly but also amplify the gains from AI adoption.[25] Deeper regional trade and economic integration could further propagate the positive spillovers from enhanced AI readiness across the region (Rayner and others 2026).

AI also poses material risks that policymakers must manage. Rapid expansion of AI-related investment may generate significant financial stability risks, particularly if adoption falls short of expectations and large-scale AI infrastructure becomes underutilized. These risks are amplified where sovereign exposures to AI-linked investments are significant, as is increasingly the case in parts of the GCC.

Beyond financial risks, AI-driven automation and skill-biased technological change pose significant medium-term labor market risks across the MENAP and CCA regions, including job displacement, widening skill mismatches, increased inequality, and associated fiscal pressures from weaker labor income tax bases and higher social spending needs. With 30–50 percent of employment exposed—the highest in GCC countries and the lowest in MENAP LICs—distributional effects may be uneven. Women, who tend to work in occupations with higher AI exposure but also stronger complementarity potential, face both risks and opportunities. Younger MENAP workers, although generally less exposed, are often concentrated in entry-level jobs with lower AI complementarity potential (Cazzaniga and others 2024; Jaumotte and others 2026), posing particular risks in economies with already high youth unemployment.

These challenges call for strengthened social protection systems and active labor market policies to support worker transitions. Education and training policies must adapt to equip workers with digital and AI-complementary skills. Sector-based training, apprenticeships, and lifelong upskilling and reskilling programs can facilitate reallocation toward new occupations and sectors (Brollo and others 2024). Given the uncertainty surrounding the pace and scale of AI diffusion—particularly generative AI—policy frameworks must remain agile. Financial regulation and fiscal policy should be prepared for a range of outcomes, including scenarios involving significant technological disruption.

[24] The World Bank's *Digital Progress and Trends Report* (World Bank 2025) notes that for low- and middle-income economies, "small AI"—defined as affordable, accessible applications designed for everyday devices—can provide an avenue for achieving immediate and tangible development impacts with limited resources. The report highlights emerging applications in sectors such as agriculture, education, and health, suggesting that such bottom-up innovation may help countries overcome infrastructure and capacity constraints and facilitate earlier adoption of AI technologies. Although potentially limited in scale, such applications may allow developing countries to bypass some traditional barriers to digital transformation.

[25] Budina and others (2023) show that prioritizing the removal of the most binding constraints on economic activity, bundling reforms in governance, business regulation, and external sector reforms, and appropriately sequencing other reforms (such as labor market and credit sector reforms) can help front-load growth gains. For emerging market and developing economies with large initial structural gaps, a comprehensive reform package is estimated to raise output by about 4 percent after two years and 8 percent after four years.

Box 3. Jordan's Progress in AI[1]

Over recent years, Jordan has strengthened its digital foundations through sustained investment in digital infrastructure and online services. By the end of 2025, 80 percent of public services had been digitized, supporting a diversified digital ecosystem—including e-commerce, fintech, govtech, and platform-based services—and reinforcing Jordan's position as a regional information and communications technology and digital services hub.

Artificial intelligence (AI) represents the next phase of Jordan's digital transformation. The country is well positioned to leverage AI to boost productivity, strengthen public service delivery, and support medium-term growth, underpinned by strong human capital and a large pool of information and communications technology and science, technology, engineering, and mathematics graduates. AI readiness has been reinforced by expanded university programs in AI and data science, national digital-skills initiatives such as *One Million Jordanian Coders* program, and strong digital government performance. This institutional capacity is reflected in Jordan's ranking in the "Very High Maturity" group of the World Bank's 2025 GovTech Maturity Index (21st globally) and 49th in the Oxford Insights Government AI Readiness Index.

AI adoption in Jordan has accelerated rapidly. The country ranks third in the Arab world and 29th globally on the AI Diffusion Index, with 25 percent of workers using AI tools in 2025—well above the global average (Microsoft AI Economy Institute 2026). Early adoption is evident across government services and the financial sector.

- *Public sector*. Jordan is deploying AI across 25 priority public projects, including water leak detection, agricultural analysis, and data-driven public services, aimed at improving efficiency and service quality in resource-constrained sectors. AI is embedded in core government systems, including tax administration, e-invoicing platforms, and AI-based audit tools. The Central Bank of Jordan has introduced a regulatory framework for AI in banking, establishing requirements for governance, model validation, risk management, data protection, transparency, and supervisory oversight. Authorities aim to expand the AI research base, support up to 50 AI start-ups, and strengthen governance through a national AI unit and Code of Ethics.
- *Private sector*. Banks, fintech firms, and IT companies are increasingly using AI in customer service, compliance, and risk management. More than 60 percent of IT firms report integrating AI into their offerings. Progress has been supported by partnerships with international organizations and development partners, including World Bank–supported programs that have expanded digital skills, firm growth, and digitization of government services beyond initial targets.

Despite this progress, AI adoption remains uneven, concentrated in large firms and public institutions. Key constraints include gaps in digital infrastructure—particularly broadband, cloud and computing access, and data availability—persistent shortages in applied AI and data science skills, and high upfront costs that limit adoption among small and medium-sized enterprises.

Addressing these constraints could help tackle structural challenges, including high youth unemployment and public-sector inefficiencies. Continued progress will require a sequenced reform agenda focused on skills development, strengthened data and AI governance, enhanced cybersecurity, and deeper public–private coordination. Sustained advances in infrastructure, human capital, governance, and financing will be essential to ensure inclusive productivity gains and a competitive, fiscally sustainable digital transformation.

[1] This box was prepared by Leen Aghabi and Sana Almunizel.

Box 4. AI as a Growth Engine in Kazakhstan[1]

Since 2024, Kazakhstan has moved beyond early digitalization initiatives, such as e-government, toward a comprehensive artificial intelligence (AI) strategy, building on the *Concept for Artificial Intelligence Development for 2024–29* (Prime Minister of the Republic of Kazakhstan 2024). The National AI Strategy will cover governance, infrastructure, human capital development, and public policy, with the objective of positioning Kazakhstan as a regional digital hub.

Kazakhstan has established a dedicated governance framework for AI, including a new law, ministry, and advisory council. The AI Law, in force since January 18, 2026, represents the base legal framework for AI, categorizing systems by risk level; setting standards for safety, transparency, and privacy; banning harmful practices; requiring content labeling; and outlining responsibilities for operators and the government in AI regulation.[2] The Ministry of Artificial Intelligence and Digital Development, established in late 2025, is responsible for implementing AI policy in priority sectors, managing the national Alem Cloud IT infrastructure, and supporting human capital development.[3] Government resolutions assign responsibilities for scaling AI solutions across ministries. The AI Advisory Council, composed of government officials, international AI experts, academics, and industry leaders, provides long-term policy guidance and promotes international collaboration.

Kazakhstan is also investing heavily in AI human capital. For 2026, the government has allocated nearly 10 billion tenge (about $23 million) to 62 AI projects implemented through research institutions and universities. The government aims to train about 1 million people in AI skills over the coming years, expanding AI literacy through programs such as AI-Sana, AI-Qyzmet, AI-People, AI Governance 500, and others, tailored for different audiences, including public servants and the general public, and supported by an increasing network of "anchor" universities offering AI courses.

Public policy aims to promote AI adoption across most economic sectors to raise productivity. In public administration, AI assistants are being deployed to support citizen services, tax administration, decision-making, and data analysis. In the extractive sector, AI is used for predictive analytics, resource monitoring, and operational automation in mineral extraction and processing. The utilities sector is developing AI tools for predictive maintenance and energy distribution efficiency. In health care, AI applications support medical diagnostics, patient management, predictive analytics, and telemedicine services.

Rapid AI adoption is expected to strengthen Kazakhstan's integration into regional and global value chains and enhance competitiveness. Recent IMF research (Cazzaniga and others 2024) finds that Kazakhstan is ahead of most emerging markets and low-income countries in terms of AI preparedness, although gaps remain relative to advanced economies, particularly in regulation and innovation. These gaps have narrowed in recent years. Kazakhstan ranked 60 out of 195 countries in the 2025 Government AI Readiness Index (Oxford Insights 2025), up from 76 the previous year, making it the leading country in Central Asia.

[1] This box was prepared by Gregorio Impavido.

[2] Law No. 230-VIII on Artificial Intelligence, Republic of Kazakhstan, accessed February 2026https://adilet.zan.kz/kaz/docs/Z2500000230.

[3] For more information on Kazakhstan's Ministry of Artificial Intelligence and Digital Development of the Republic of Kazakhstan (n.d.).

Box 5. Saudi Arabia's AI Ambitions: Strategies and Challenges[1]

Positioning Saudi Arabia at the forefront of global artificial intelligence (AI) adoption and development is a central pillar of Vision 2030. This ambition builds on significant digital transformation in recent years,[2] including strengthened digital infrastructure, enhanced regulation, and a growing ICT workforce. Within this context, Saudi Arabia's National AI Strategy aims to attract investment, expand AI talent, establish an enabling regulatory environment, foster innovation, and scale AI applications across the economy.[3] The authorities also aim to increase exports of data and ICT-related services, including through the development of large-scale AI data centers. Whereas the initial focus has been on meeting rising domestic demand, the strategy envisages a shift toward serving international clients by 2030, particularly in the United States.

The strategy is characterized by strong state leadership and the mobilization of sovereign capital, with emphasis on building foundational infrastructure alongside scaling applications. Institutionally, the Saudi Data and Artificial Intelligence Authority leads strategy design and governance, and the Digital Government Authority drives AI adoption across public entities. The Public Investment Fund plays a catalytic role in developing the private AI ecosystem while seeking to crowd in private investment.

Saudi Arabia's public sector has been at the forefront of AI adoption, supported by sustained investment in digital government and strong global benchmark performance. AI is increasingly embedded across ministries, improving service delivery, while also catalyzing private sector uptake. Across health care, education, and finance, applications include remote diagnostics, adaptive learning platforms, advanced analytics, fraud detection, risk assessment, and support for fintech and digital payments.

Saudi Arabia largely surpassed its 2025 interim AI targets. For example, the AI workforce reached 10,900, whereas cumulative local investment and foreign direct investment in data and AI totaled $132 billion during 2020–25. The institutional framework for data governance and cross-sector AI integration has also strengthened.

Despite this progress, key constraints remain. Shortages persist in advanced AI research and engineering talent, reflecting still-maturing domestic training pipelines, intense competition from global technology hubs, and visa-related frictions. Private sector participation remains limited, with investment dominated by the state and Public Investment Fund amid uncertain commercial returns. Data governance frameworks require further clarity, particularly for cross-border data flows, to enhance regulatory certainty and attract foreign partners. Execution risks are also increasing as large-scale AI and giga projects place pressure on sovereign balance sheets, bank funding, energy supply, and inter-agency coordination.

Going forward, Saudi Arabia will need to remain adaptive to technological change, global competitive pressures, and evolving regulation to ensure that AI investment delivers broad-based productivity gains and supports economic diversification.

[1] This box was prepared by Yuan Gao Rollinson.

[2] See Box 1 of Bi and others (2025).

[3] *National Strategy for Data and AI* (adopted June 2020), Saudi Data and Artificial Intelligence Authority (SDAIA), https://sdaia.gov.sa/en/SDAIA/SdaiaStrategies/Pages/NationalStrategyForDataAndAI.aspx.

Annex 1. Data and Model Details

Annex Table 1.1. Labor Market Data for AI Exposure Calculations

Model Bloc	Country	Occupation Data	Industry Data
United Arab Emirates	United Arab Emirates	2-digit occupation data from ILO.	2-digit industry data from ILO.
Saudi Arabia	Saudi Arabia	Applied UAE occupation-industry mapping to compute country-specific occupational distribution within each industry.	2-digit industry codes from Haver.
The rest of the GCC	Bahrain	Applied UAE occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied UAE-SAU average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.
	Kuwait	Applied UAE occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied UAE-SAU average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.
	Oman	Applied UAE occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied UAE-SAU average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.
	Qatar	Applied UAE occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied UAE-SAU average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.
MENAP EMMI	Algeria	Applied EGY-IRN-IRQ-JOR-LBN-PAK-TUN occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied EGY-IRN-IRQ-JOR-LBN-PAK-TUN average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.
	Egypt	2-digit occupation data from ILO.	2-digit industry data from ILO.
	Iran	2-digit occupation data from ILO.	2-digit industry data from ILO.
	Iraq	2-digit occupation data from ILO.	2-digit industry data from ILO.
	Jordan	2-digit occupation data from ILO.	2-digit industry data from ILO.
	Lebanon	2-digit occupation data from ILO.	2-digit industry data from ILO.
	Libya	Applied EGY-IRN-IRQ-JOR-LBN-PAK-TUN occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied EGY-IRN-IRQ-JOR-LBN-PAK-TUN average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.

(*continued*)

Annex Table 1.1. Labor Market Data for AI Exposure Calculations (continued)

Model Bloc	Country	Occupation Data	Industry Data
	Morocco	Applied EGY-IRN-IRQ-JOR-LBN-PAK-TUN occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied EGY-IRN-IRQ-JOR-LBN-PAK-TUN average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.
	Pakistan	2-digit occupation data from ILO.	2-digit industry data from ILO.
	Tunisia	2-digit occupation data from ILO.	2-digit industry data from ILO.
	West Bank & Gaza	Applied EGY-IRN-IRQ-JOR-LBN-PAK-TUN occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied EGY-IRN-IRQ-JOR-LBN-PAK-TUN average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.
MENAP LIC	Afghanistan	2-digit occupation data from ILO	2-digit industry data from ILO
	Djibouti	Applied AFG-SOM-SDN occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied AFG-SOM-SDN average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.
	Mauritania	Applied AFG-SOM-SDN occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied AFG-SOM-SDN average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.
	Somalia	2-digit occupation data from ILO	2-digit industry data from ILO
	Sudan	2-digit occupation data from ILO	2-digit industry data from ILO
	Syria	Applied AFG-SOM-SDN occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied AFG-SOM-SDN average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.
	Yemen	Applied AFG-SOM-SDN occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied AFG-SOM-SDN average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.
CCA	Armenia	Applied GEO-KGZ occupation-industry mapping to compute country-specific occupational distribution within each industry.	2-digit industry data from ILO
	Azerbaijan	Applied GEO-KGZ occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied GEO-KGZ average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.
	Georgia	2-digit occupation data from ILO.	2-digit industry data from ILO.
	Kazakhstan	Applied GEO-KGZ occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied GEO-KGZ average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.

(*continued*)

Annex Table 1.1. Labor Market Data for AI Exposure Calculations (continued)

Model Bloc	Country	Occupation Data	Industry Data
	Kyrgyz Republic	2-digit occupation data from ILO.	2-digit industry data from ILO.
	Tajikistan	Applied GEO-KGZ occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied GEO-KGZ average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.
	Uzbekistan	Applied GEO-KGZ occupation-industry mapping to compute country-specific occupational distribution within each industry.	Applied GEO-KGZ average (2-digit industry data) to disaggregate country-specific 1-digit industry data from ILO.

Source: IMF staff calculations.
Note: CCA = Caucasus and Central Asia; EMMI = emerging market and middle-income economy; GCC = Gulf Cooperation Council; ILO = International Labour Organization; LIC = low-income country; MENAP = Middle East, North Africa, Afghanistan, and Pakistan.

Annex Table 1.2. MENAP and CCA Country Sample Coverage

Model Bloc	Country Code	Country	Income Group
Saudi Arabia	SAU	Saudi Arabia	EMMI
United Arab Emirates	UAE	United Arab Emirates	EMMI
Rest of the GCC	BHR	Bahrain	EMMI
	KWT	Kuwait	EMMI
	OMN	Oman	EMMI
	QAT	Qatar	EMMI
CCA	ARM	Armenia	EMMI
	AZE	Azerbaijan	EMMI
	GEO	Georgia	EMMI
	KAZ	Kazakhstan	EMMI
	KGZ	Kyrgyz Republic	LIC
	TJK	Tajikistan	LIC
	TKM	Turkmenistan	EMMI
	UZB	Uzbekistan	LIC
MENAP EMMI	ALG	Algeria	EMMI
	EGY	Egypt	EMMI
	IRN	Iran	EMMI
	IRQ	Iraq	EMMI
	JOR	Jordan	EMMI

(*continued*)

Annex Table 1.2. MENAP and CCA Country Sample Coverage (continued)

Model Bloc	Country Code	Country	Income Group
	MAR	Morocco	EMMI
	LBN	Lebanon	EMMI
	LBY	Libya	EMMI
	PAK	Pakistan	EMMI
	TUN	Tunisia	EMMI
	WBG	West Bank and Gaza	EMMI
MENAP LIC	AFG	Afghanistan	LIC
	DJI	Djibouti	LIC
	MRT	Mauritania	LIC
	SOM	Somalia	LIC
	SDN	Sudan	LIC
	SYR	Syrian Arab Republic	LIC
	YEM	Yemen	LIC

Source: IMF staff calculations.

Note: The model also includes two additional blocs: advanced economies and the rest of the world not covered by other blocs. CCA = Caucasus and Central Asia; EMMI = emerging market and middle-income economy; GCC = Gulf Cooperation Council; LIC = low-income country; MENAP = Middle East, North Africa, Afghanistan, and Pakistan.

Annex Table 1.3. Aggregation of Industries in GIMF

Sector	ISIC2 Code	Description
AI-intensive	D21	Pharmaceuticals, medicinal chemical and botanical products
AI-intensive	D26	Computer, electronic, and optical equipment
AI-intensive	D51	Air transport
AI-intensive	D58T60	Publishing, audiovisual, and broadcasting activities
AI-intensive	D61	Telecommunications
AI-intensive	D62T63	IT and other information services
AI-intensive	D64T66	Financial and insurance activities
AI-intensive	D69T75	Professional, scientific, and technical activities
Tradable	D01T02	Agriculture, hunting, and forestry
Tradable	D03	Fishing and aquaculture
Tradable	D05T06	Mining and quarrying; energy-producing products
Tradable	D07T08	Mining and quarrying, non-energy-producing products
Tradable	D09	Mining support service activities
Tradable	D10T12	Food products, beverages, and tobacco

(*continued*)

Annex Table 1.3. Aggregation of Industries in GIMF (continued)

Sector	ISIC2 Code	Description
Tradable	D13T15	Textiles, textile products, leather, and footwear
Tradable	D16	Wood and products of wood and cork
Tradable	D17T18	Paper products and printing
Tradable	D19	Coke and refined petroleum products
Tradable	D20	Chemical and chemical products
Tradable	D22	Rubber and plastic products
Tradable	D23	Other non-metallic mineral products
Tradable	D24	Basic metals
Tradable	D25	Fabricated metal products
Tradable	D27	Electrical equipment
Tradable	D28	Machinery and equipment, nec
Tradable	D29	Motor vehicles, trailers, and semi-trailers
Tradable	D30	Other transport equipment
Tradable	D31T33	Manufacturing nec; repair and installation of machinery and equipment
Tradable	D49	Land transport and transport through pipelines
Tradable	D50	Water transport
Tradable	D52	Warehousing and support activities for transportation
Tradable	D55T56	Accommodation and food service activities
Non-tradable	D35	Electricity, gas, steam, and air conditioning supply
Non-tradable	D36T39	Water supply; sewage, waste management, and remediation activities
Non-tradable	D41T43	Construction
Non-tradable	D45T47	Wholesale and retail trade; repair of motor vehicles
Non-tradable	D53	Postal and courier activities
Non-tradable	D68	Real estate activities
Non-tradable	D77T82	Administrative and support services
Non-tradable	D84	Public administration and defense; compulsory social security
Non-tradable	D85	Education
Non-tradable	D86T88	Human health and social work activities
Non-tradable	D90T93	Arts, entertainment, and recreation
Non-tradable	D94T96	Other service activities
Non-tradable	D97T98	Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use

Source: Cerutti and others (2025).
Note: AI = artificial intelligence; GIMF = Global Integrated Monetary and Fiscal Model.

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